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MM6108 Hardware Design Guide

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1. Introduction

The following information can be used as a guide on how to design circuits using the MM6108. It will cover the recommended pin connections, supporting circuitry and various RF front end architectures.

2. System Planning & Preparation

Time to market can be shortened by sequencing tasks so that items with upstream dependencies are started early. When fundamental choices around components, layout constraints, and system requirements are changed or deferred to later in the project, they can lead to redesigns or delays. Resolving these upfront means work can continue without being blocked by missing information.

2.1. Impedance Control

Contact the PCB manufacturer early in the design process to obtain their recommended coplanar waveguide trace geometry based on the dielectric constant, loss tangent, and dielectric thickness associated with their process. The MM6108 is designed for 0.2 mm wide pads with 0.2 mm spacing, which makes a 50 Ω characteristic impedance difficult to achieve on a 2-layer board; a closely spaced ground plane is needed, necessitating at least a 4-layer stackup. For FR-4 fiberglass substrate, the thickness of the first dielectric layer should be in the order of 0.1 mm.

PCB trace geometry is often limited by the smallest-pitch component in the RF chain. It is advisable to identify these key RF components early and base the impedance-controlled layout around them. Components and connectors with pad widths near 0.2 mm should be preferred, as they help minimize impedance mismatches caused by trace-width variation.

Impedance calculation tools and formula-based estimators may be useful for approximating trace dimensions. However, these models are not necessarily derived from exact field solutions; many are empirical or quasi-empirical curve fits to simulation and measurement data. Manufacturers typically apply their own field solvers and impedance tuning, so collaboration is essential. Coordination with your PCB manufacturer should ideally take place before layout begins; otherwise, you may be forced to redo routing if trace geometries must be adjusted later. When using multiple vendors, an equal number of unique trace geometries will be required to accommodate each vendor.

Advanced simulation tools (e.g. full-wave 3D solvers) offer greater accuracy than equation-based methods by accounting for geometry, material variation, and board-specific features. However, this level of modeling is typically only warranted when losses or impedance variations of less than a decibel are significant to the application.

2.2. Component Availability

Verify the long-term availability and lead times for critical components early in the design process. This ensures your product's lifecycle won't be jeopardized by supply issues or sudden obsolescence. Example component lists are provided in Section 6 of this guide. They may be used as proxies for budgetary purposes or followed directly if one of the example designs matches your application.

It can be helpful to address availability risks for rare or non-substitutable parts early in the design process, while deferring broader component ordering until after the PCB has been released for fabrication. This approach is applicable when component lead times are shorter than PCB turnaround, allowing part procurement to proceed in parallel with PCB fabrication.

2.3. Antenna Selection

The antenna will both influence and be influenced by the final enclosure design, which—at the time of hardware design—is likely still evolving. Antenna performance depends heavily on its immediate surroundings, including the enclosure, internal cabling, external cabling, nearby metallic structures (such as the PCB), and objects in the deployment environment. These factors should be kept in mind throughout the design process to avoid costly redesigns if performance issues are discovered late.

It is advisable to obtain antenna samples and begin testing early, even while PCB design is underway. Return loss and radiation pattern (gain versus angle) should be measured using a setup that approximates the final enclosure and cabling. Be aware that changes to the enclosure, cables, or surrounding environment may force a change in antenna selection—yet those elements may themselves depend on a PCB layout that is still in development. This interdependency must be managed carefully throughout development.

2.3.1. Antenna Connectors

If the product interfaces to a standard 50 Ω RF connector (as opposed to an integrated PCB antenna), there are no specific tasks that must be completed before design commences. The appropriate focus for the PCB designer is to maintain a consistent

50 Ω impedance throughout the RF path using best-practice layout techniques. System-level testing still needs to be considered early, as interactions between the antenna, enclosure, cabling, and nearby conductive structures will influence performance and affect final choices around antenna type, connector placement, and enclosure design.

2.3.2. Off-The-Shelf Antennas

When selecting an off-the-shelf antenna for a Wi-Fi HaLow (IEEE 802.11ah) module, it is critical to ensure compatibility with the sub-GHz frequency range (typically 860–930 MHz, depending on region) with a bandwidth of at least 8MHz. Wi-Fi HaLow operates on wider bandwidths than other common ISM-band technologies (such as LoRa), so care must be taken to check the operating bandwidth if the antenna is advertised for use with one of these other technologies. For general-purpose deployment and consistent coverage, a dipole antenna is recommended due to its omnidirectional radiation pattern in the azimuth plane. This facilitates uniform signal strength around the antenna, ideal for applications where client devices may be distributed radially. Key parameters to consider include antenna gain (typically 1–3 dBi for dipoles), connector type (must match the end-product's or module's RF interface), mechanical form factor, and environmental rating.

In addition to performance, regulatory compliance must be considered when selecting an antenna for a Wi-Fi HaLow module. The chosen antenna must undergo and pass regulatory testing as part of the module and end-product certification process, including assessments for radiated emissions, spurious emissions, and effective isotropic radiated power (EIRP) to ensure conformance with regional standards (e.g., FCC Part 15.247 in the US or ETSI EN 300 220 in Europe). The antenna used during these tests must be explicitly listed in the regulatory filings. Consequently, only antennas that have been tested and approved as part of the certification process are permitted for use. Any change in antenna type, gain, or model post-certification may necessitate either re-testing or a permissive change filing with the relevant authority.

For Morse Micro reference designs, the Pulse Larsen W1063 antenna has been validated in both testing and regulatory certification purposes. This antenna is a dipole with a nominal gain of 1 dBi, making it well-suited for omnidirectional applications within the sub-GHz band. Customers integrating Morse Micro's Wi-Fi HaLow modules may leverage this antenna, or any alternative antenna with equivalent type and gain characteristics, to file a self-declared Class I permissive change under applicable regulatory frameworks. However, if a different antenna type is used—such as a monopole, patch, or a model with higher gain—this constitutes a Class II permissive change. In such cases, additional regulatory testing is required, including radiated power, radiated spurious emissions, and EIRP spot checks, to validate continued compliance. The alternative antenna must then be explicitly listed in the updated regulatory documentation to ensure that the end product remains certified.

It is recommended to evaluate antenna performance in a radiofrequency anechoic chamber using the final product assembly, including all enclosures, cables, and connectors. This approach provides an accurate characterization of the antenna's radiation pattern, gain, and efficiency as integrated in the end device. Testing in the final mechanical configuration helps identify detuning effects, coupling, or obstructions that may degrade RF performance, ensuring the antenna operates as intended in the real deployment environment. This data is also valuable for validating compliance with regulatory requirements and optimizing system performance.

2.3.3. Electrically Small Antennas

At 915 MHz, antennas shorter than approximately 30 mm are considered electrically small. Designs in this range are especially difficult, with significant trade-offs in efficiency, bandwidth, gain, and susceptibility to detuning from nearby materials.

More generally, any antenna smaller than a quarter wavelength (approximately 80 mm) should be approached with caution. As physical size decreases below this point, impedance matching becomes more sensitive, performance less predictable, and more dependent on enclosure layout and nearby conductors. Early validation and testing are strongly recommended for antennas below this threshold. Plan for extensive impedance matching, validation, and possibly iterative tuning of the matching network and physical layout to ensure acceptable performance.

2.3.4. PCB Antennas

The performance of a PCB antenna depends on the manufacturing process, PCB material, enclosure, cabling, and other nearby structures. Design parameters—such as the required gain for the PA and LNA, and the recommended matching topology—are best determined with knowledge of the antenna's impedance and radiation characteristics.

It is recommended to plan to design and build early prototypes specifically for antenna evaluation, even before laying out the full product. These can include the enclosure, dummy connectors, cabling, and the antenna section, while omitting other circuitry. Such setups allow return-loss and pattern measurements that place the design in the right ballpark. While further adjustments may be expected, this process reduces uncertainty and enables more informed decisions before significant investment in the final layout.

3. Chip Pinout

This section provides the details needed to create a footprint and schematic symbol for the device. It includes a visual overview of the QFN48 package and a table of pin functions, covering physical locations, electrical characteristics, and signal definitions.

3.1. Component Land Pattern

Rounded pads of uniform size and spacing are recommended.

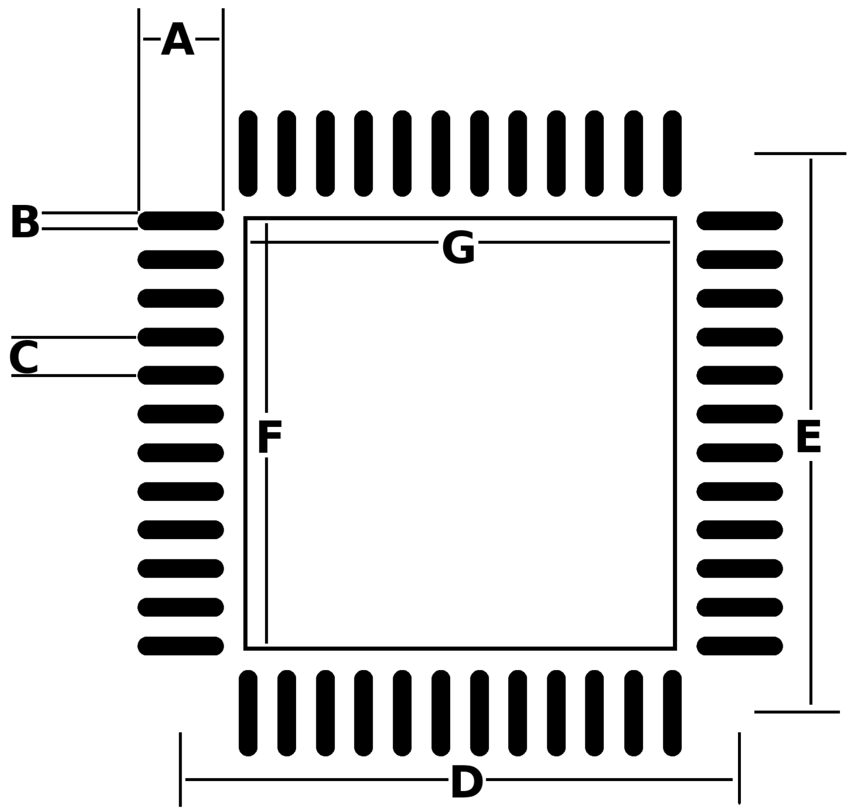


Figure 1. Land pattern for QFN48 package

Dimensions are provided in mm. Alternative dimensions in mil (thou) have been provided for convenience. Where ambiguity exists (i.e. due to rounding) the mm dimensions should be treated as the source of truth.

Key	Dimension (mm)	Dimension (mil)
A	0.90	35.5
B	0.20	7.90
C	0.40	15.8
D	0.58	22.8
E	5.80	228
F	4.50	177
G	4.50	177

Table 1. Key Dimensions associated with Figure 1.

3.2. Solder Stencil Pattern

For pick and place assembly, the following is recommended for the solder paste layer.

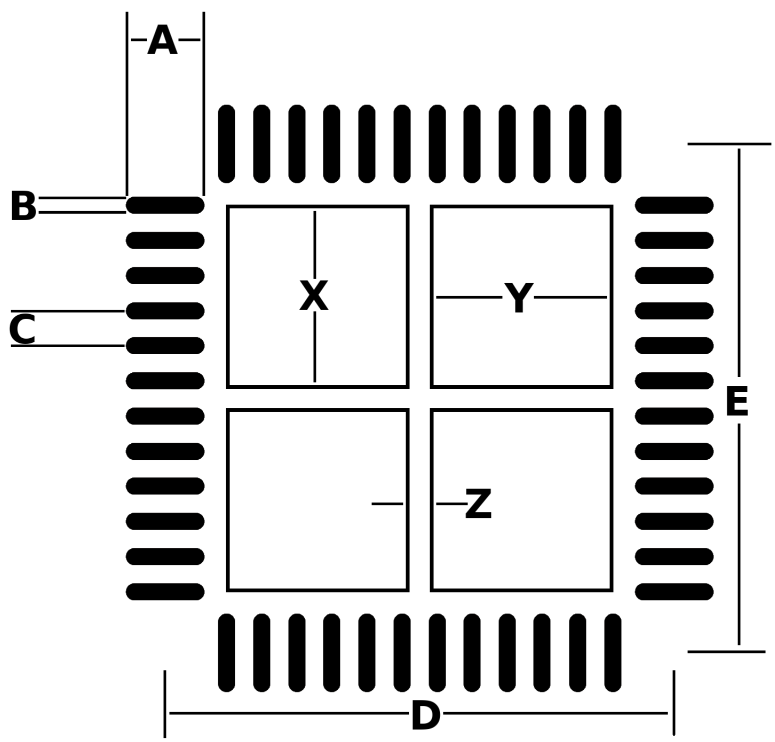


Figure 2. Stencil dimensions for QFN48 package

As before, the associated dimensions are provided in the following table:

Key	Dimension (mm)	Dimension (mil)
A	0.90	35.5
B	0.20	7.90
C	0.40	15.8
D	0.58	22.8
E	5.80	228
F	4.50	177
G	4.50	177

Table 2. Key Dimensions associated with Figure 2.

3.3. Schematic Symbol

The diagram below shows a top view of the device package with each pin numbered and labeled. Pin 1 is indicated by a dot, and the remaining pins are numbered counterclockwise around the package. Signal names are annotated alongside each pin.

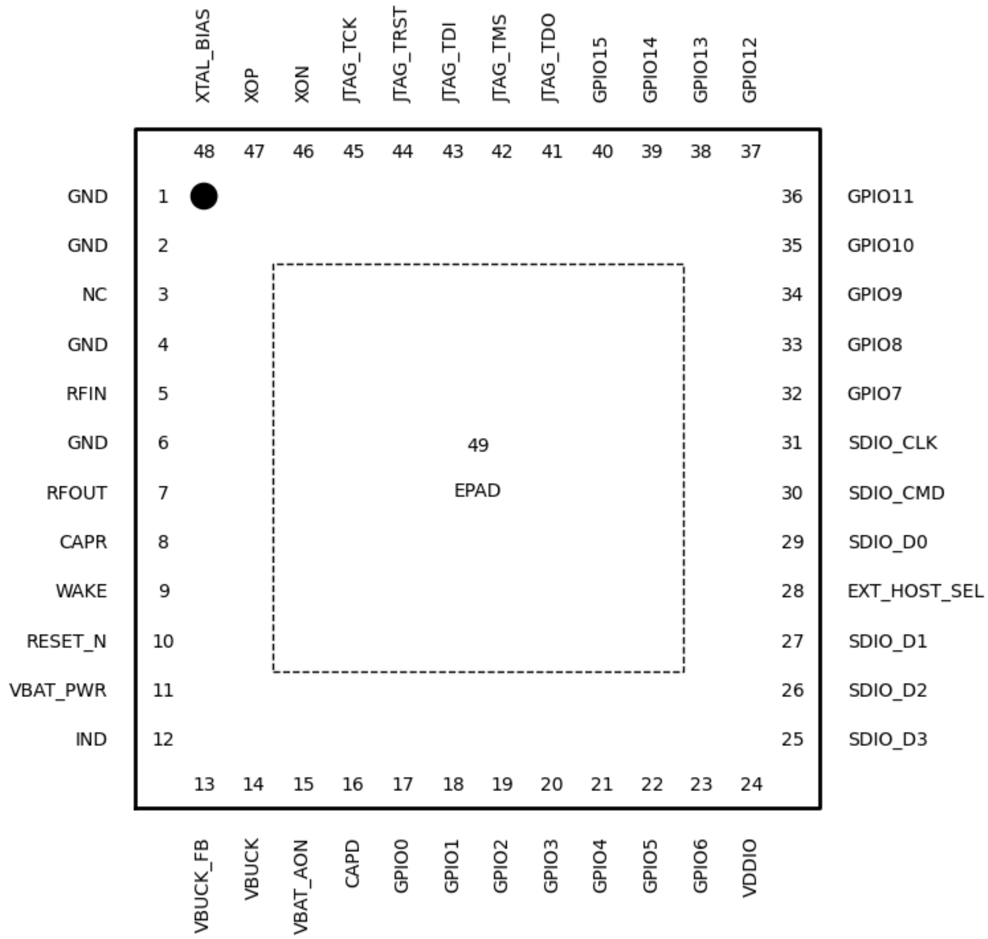


Figure 3. Diagram showing pinout information for MM6108

The table below lists all package pins in numerical order, along with their signal names, electrical types, and functional descriptions. The "Type" column identifies each pin’s role, such as analog interface or ground return, to support interpretation during circuit design and layout. Descriptions are provided in the “Function” column for clarity, especially in cases where the signal name alone may not convey the intended usage.

Number	Name	Type	Function
1	GND	Ground	Ground
2	GND	Ground	Ground
3	NC	-	Leave unconnected

Number	Name	Type	Function
4	GND	Ground	Ground
5	RFIN	Analog	Wi-Fi Radio RX Input
6	GND	Ground	Ground
7	RFOUT	Analog	Wi-Fi Radio TX Output
8	CAPR	Power	Internal LDO bypass capacitor
9	WAKE	Input	Input signal to bring the MM6108 out of deep sleep and snooze modes
10	RESET_N	Input	Active Low Reset
11	VBAT_PWR	Power	Main regulator supply input
12	IND	Output	Buck regulator switching node
13	VBUCK_FB	Input	Sense feedback for internal BUCK regulator
14	VBUCK	Power	Input for all internal LDO regulators
15	VBAT_AON	Power	Power rail for always-on logic
16	CAPD	Power	Internal LDO bypass capacitor
17	GPIO0	Output	Output signal from MM6108 to host to indicate busy state
18	GPIO1	I/O	General Purpose IO
19	GPIO2	I/O	General Purpose IO
20	GPIO3	I/O	General Purpose IO
21	GPIO4	I/O	General Purpose IO
22	GPIO5	I/O	General Purpose IO
23	GPIO6	I/O	General Purpose IO
24	VDDIO	Power	VDD supply for digital IO
25	SDIO_D3	I/O	SDIO D3 / SPI CS
26	SDIO_D2	I/O	SDIO D2
27	SDIO_D1	I/O	SDIO D1 / SPI INT

Number	Name	Type	Function
28	EXT_HOST_SEL	Input	Boot Mode
29	SDIO_D0	I/O	SDIO D0 / SPI MISO
30	SDIO_CMD	I/O	SDIO CMD / SPI MOSI
31	SDIO_CLK	I/O	SDIO CLK / SPI CLK
32	GPIO7	I/O	General Purpose IO
33	GPIO8	I/O	General Purpose IO
34	GPIO9	I/O	General Purpose IO
35	GPIO10	I/O	General Purpose IO
36	GPIO11	I/O	General Purpose IO
37	GPIO12	I/O	General Purpose IO
38	GPIO13	I/O	General Purpose IO
39	GPIO14	I/O	General Purpose IO
40	GPIO15	I/O	General Purpose IO
41	JTAG_TDO	Output	JTAG Data Out
42	JTAG_TMS	Input	JTAG Mode Select
43	JTAG_TDI	Input	JTAG Data In
44	JTAG_TRST	Input	JTAG Test Reset
45	JTAG_TCK	Input	JTAG Test Clock
46	XON	Output	Crystal Output
47	XOP	Input	Crystal Input
48	XTAL_BIAS	Power	Decoupling capacitor for the crystal oscillator power rail
49	EPAD	Power	Ground

Table 3. Description of pins number and function for MM6108

4. Schematic Design

An example schematic design is shown below. It includes all critical elements of a typical MM6108 design such as power supply decoupling, external components for the buck converter and the crystal oscillator.

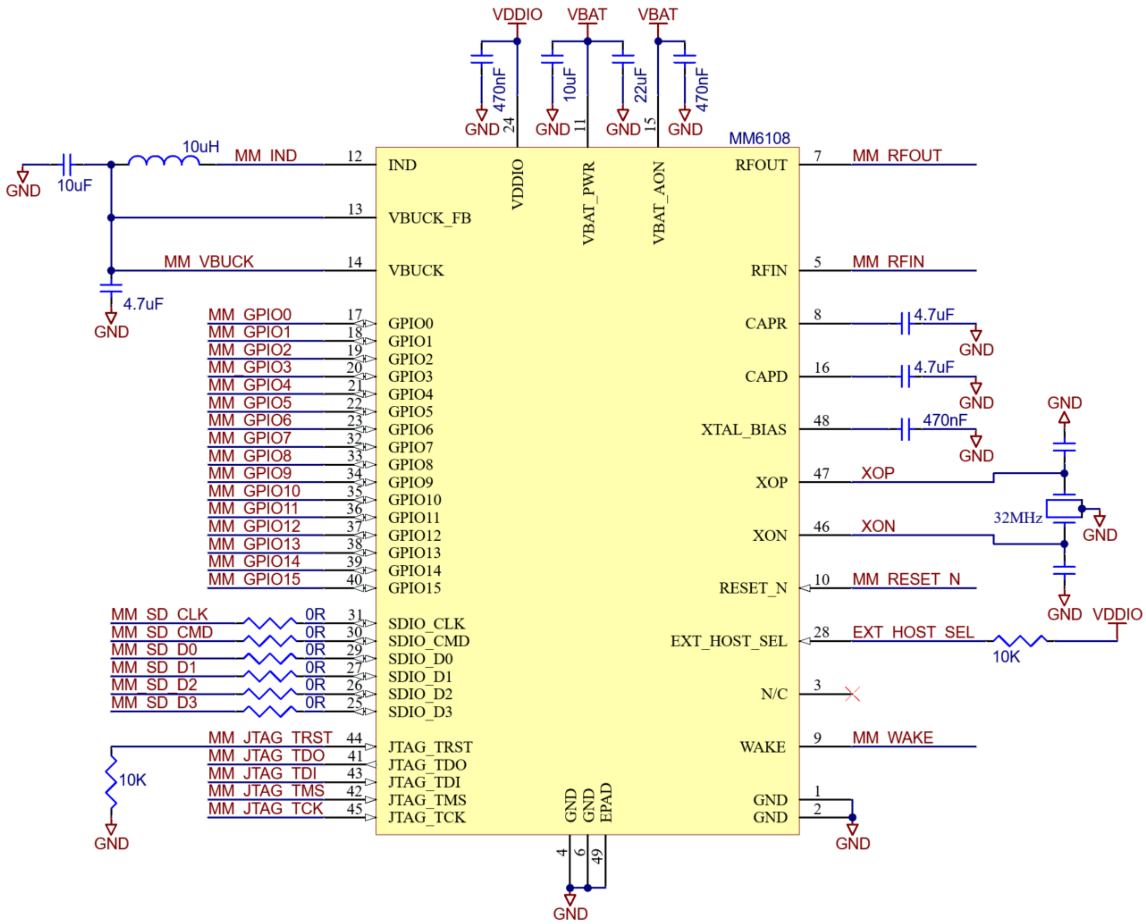


Figure 4. Typical application schematic for the MM6108 and surrounding circuitry

A recommended schematic design workflow is to begin with the RF signal path, followed by the power supply and clocking circuits. Once those are in place, complete the decoupling networks, I/O configuration, diagnostics, and protection circuitry.

4.1. Radio Frequency

The RF section should be the primary design consideration. It is the core function of the device and one of the more challenging aspects of the design.

4.1.1. Impedance Control

RFIN, RFOUT, and similar nets should be designated as impedance-controlled (50 Ω characteristic impedance) in the schematic editor to ensure the correct routing constraints are applied during layout.

4.1.2. Antenna and Associated Matching

If an off-the-shelf 50 Ω antenna is used (or if the RF output is simply routed to a 50 Ω connector to support standard antenna types) then matching is not required; the best performance is achieved by routing directly, with a short trace from the RF output to the connector. However, many designers choose to include a placeholder pi or tee network. This introduces a performance trade-off (wider pads, extra trace length and additional components leading to added mismatch, loss and cost) in exchange for the flexibility that if issues arise later in the development process, these footprints may allow for filtering or attenuation.

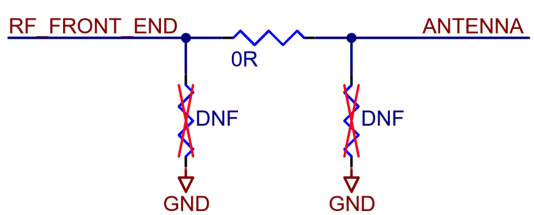


Figure 5. A pi network

If the antenna is **not** 50 Ω (as is often the case with electrically short PCB antennas) then impedance matching becomes mandatory. A generic pi or tee network can be included to facilitate matching, which poses minimal effort for a roughly matched system. However, the optimal matching circuitry will depend on the exact impedance presented by the antenna, including the PCB, inside the enclosure, and deployed in a representative scenario. VNA measurements of antenna-only prototypes provide the data needed to implement optimized front-end matching.

Antenna Design Considerations: When designing a small-form-factor PCB antenna, changes in the antenna geometry will affect the load presented by the (unmatched) antenna. To the extent this may be influenced, the top region of the Smith chart is recommended (Figure 6), as it may be matched using purely capacitive matching, which behaves more ideally (exhibiting less loss and magnetic coupling) than inductive matching.

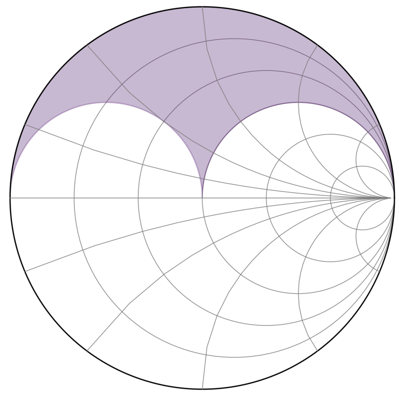


Figure 6. Target impedance region on the Smith chart for small-form-factor PCB antennas

4.2. RF Front End

The MM6108 provides a transmit output of up to 8 dBm and can receive signals as low as −107 dBm (at MCS10). To operate with a shared antenna and/or achieve improved system-level performance, external front-end circuitry may be employed. This typically includes an RF switch to select between transmit and receive paths, and may also include a low-noise amplifier (LNA) to improve receive sensitivity or a power amplifier (PA) to extend transmit range. Some RF front end design examples are provided in Section 6.

4.3. Buck Converter

The MM6108 includes an integrated buck converter that provides power to internal regulators for the RF, analog, and digital domains. This buck converter requires a single external inductor and two output capacitors.

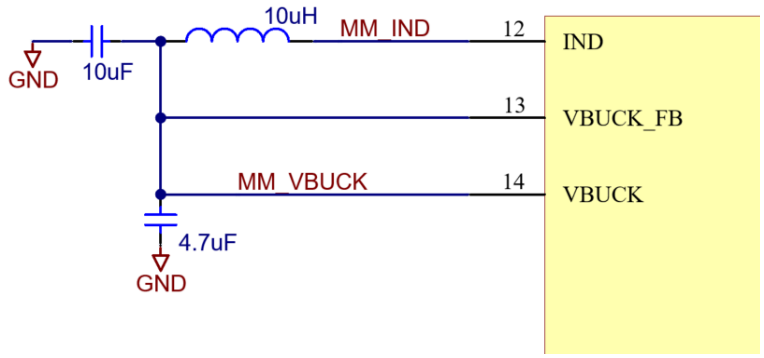


Figure 7. Schematic of the buck converter external components

The inductor (connected between the IND and VBUCK pins) should have a value of 10 uH (±20%), a saturation current above 1.2 A and an ESR less than 500 m Ω (lower ESR can further improve efficiency). Magnetically shielded inductors are recommended to limit electromagnetic interference and reduce magnetic coupling into nearby traces, crystals, or antennas, especially in compact or RF-sensitive layouts.

The first capacitor (routed closer to the inductor) must be 10 µF, ceramic type (X5R), and rated to 10 V. The second capacitor (routed closer to VBUCK) must be 4.7 µF, ceramic type (X5R), and rated to 6.3 V.

4.4. Power Supply

The performance of the MM6108 depends critically on the quality and stability of its power supply. While the device includes internal regulators to generate the required internal voltages, it is sensitive to variations, ripple, and noise on the primary voltage supply.

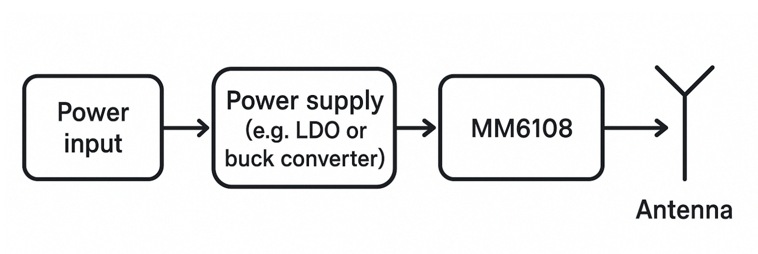


Figure 8. Simplified system block diagram showing power path

The output should remain within the 3.0–3.6 V range across all load conditions (from sleep currents to peak TX current) and under expected variations in input supply. Gradual shifts in power supply voltage (e.g. from a battery discharging) are tolerated by the internal LDO and buck converters. However, high-frequency noise on the supply remains a critical consideration for RF performance.

The MM6108 is typically powered by an external voltage regulator, which should have an output ripple below 30 mV peak-to-peak, supply at least 500 mA continuously, and support short bursts up to 1.5 A during wake-up and power-on. If a switch-mode power supply is used, it is recommended to include a footprint for an RF shielding can that fully encloses its perimeter. This allows for fitting a shield if required to suppress radiated emissions and preserve receiver sensitivity. Pay close attention to the effective series resistance (ESR) in the supply path, which combines with inrush currents to produce voltage dips. For example, a 200 mΩ ESR can drop a 3.3 V rail to 3.0 V under such conditions, while a 50 mΩ ESR would limit the drop to about 3.25 V.

Ensure that VDDIO is never greater than VBAT at any time. It is the responsibility of the system power architecture to guarantee this constraint under all operating conditions, including during power-up, power-down, and any transient scenarios. Special attention should be given to scenarios where VBAT may be unintentionally disabled or lost, to ensure that VDDIO is either sequenced off first or remains safely clamped below VBAT.

4.5. Reference Crystal

A 32 MHz crystal with an equivalent series resistance (ESR) of 50 Ω or less is required. The Siward XTL901-M202-001 has been validated with the MM6108 and is recommended for designs requiring a known-compatible part. The total frequency stability of the selected crystal (including initial tolerance, temperature variation, and aging) must remain within ±20 parts per million (ppm) over the full operating temperature range of –40 °C to +85 °C.

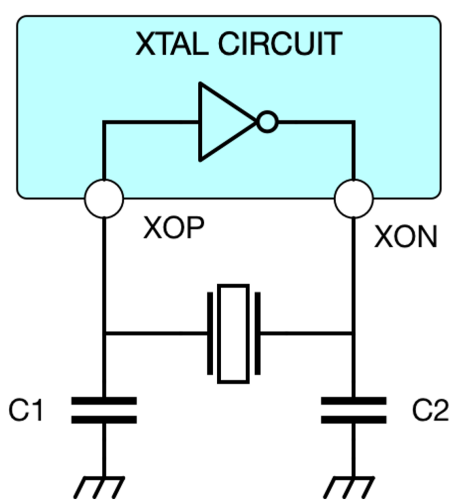


Figure 9. A crystal resonator circuit with crystal and load capacitors shown

Two load capacitors are generally required (one on each crystal pin), the value of which depends on the load capacitance (C_L) specified by the crystal manufacturer. Stray capacitance (C_{stray}) is typically in the range of 2 to 3 pF, though exact values are dependent on the layout geometry. Capacitors (C_1=C_2=C_x) should use NP0 or C0G ceramic dielectric for stable performance over temperature, and must be rated for at least 6.3 V. These capacitors must be placed as close as possible to the crystal pins to minimize series inductance and preserve layout symmetry.

$$C_L = \frac{1}{2} C_x + C_{stray}$$

This formula provides a solid starting point for selecting capacitor values; however, it is recommended to verify and fine-tune these values during initial hardware bring-up. Once prototypes are available, measuring key performance metrics (such as frequency offset and EVM) while adjusting the capacitor values in the surrounding range helps account for layout-specific parasitics and ensures optimal performance. Note that the final values of C_1 and C_2 need not necessarily be exactly the same - the final load capacitance is the sum of these two capacitances plus the total parasitic capacitance (C_{stray}).

4.6. Decoupling

4.6.1. Nominal Decoupling Capacitor Values

The CAPR and CAPD pins are each decoupled with a 4.7 µF ceramic capacitor, rated for at least 4 V. The XTAL_BIAS pin must be decoupled with a 470 nF ceramic capacitor, rated for at least 4 V. These pins are internal LDO bypass outputs and must not be used to power any external circuitry.

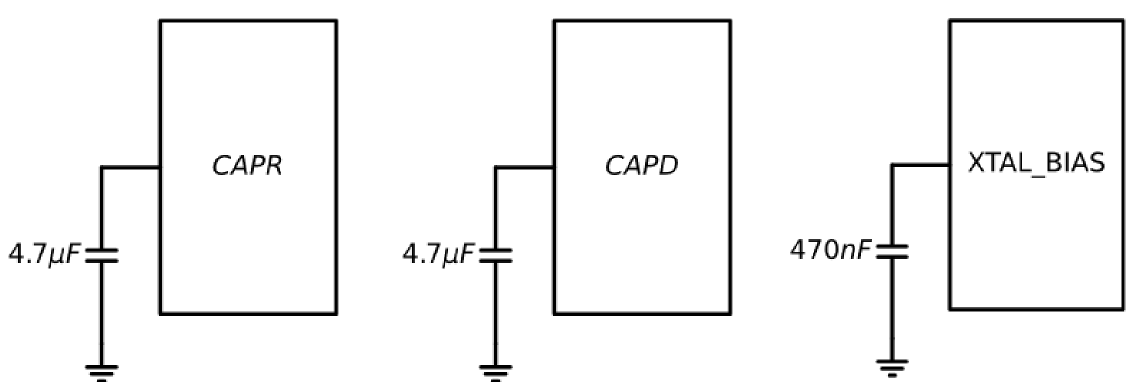


Figure 10. Decoupling capacitors required for internal low dropout regulators

The VBAT_AON pin must be decoupled with a 470 nF ceramic capacitor, rated for at least 6.3 V, and placed close to the pin. The VDDIO pin must be decoupled with a 470 nF ceramic capacitor, rated for at least 6.3 V.

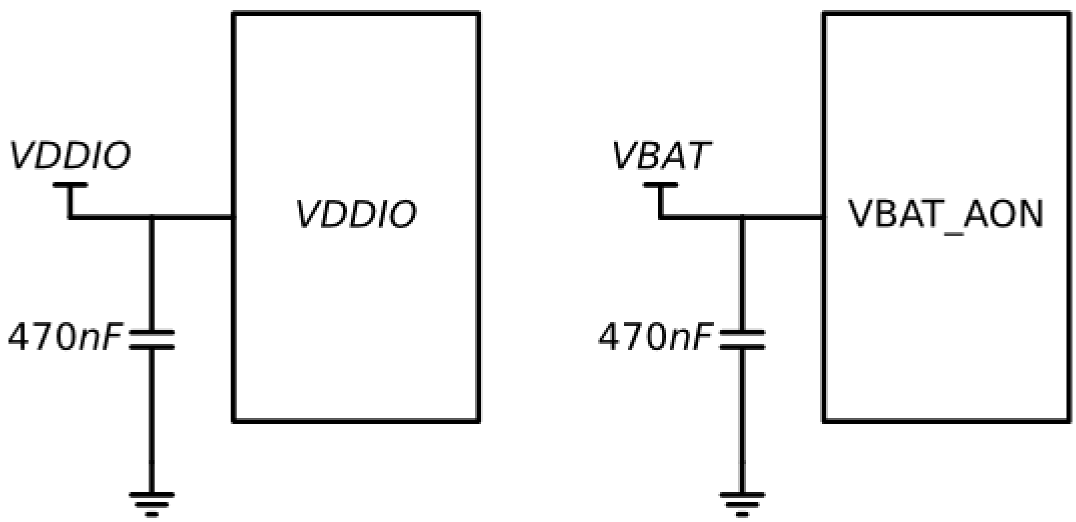


Figure 11. Decoupling capacitors required for main voltage supplies VDDIO and VBAT_AON

Note that sudden spikes of current may occur upon wake-up to charge the internal regulators, which has implications for the decoupling capacitance required at VBAT_PWR. Approximately 18 µJ of energy will be required to wake up the device, which means at least 32 µF of total decoupling capacitance should be provided to support this energy demand. At least 10 µF is recommended close to the device with the remainder being supplied upstream (but before the net is shared with other devices). Even with this bulk capacitance, momentary voltage dips in a 3.3 V supply as low as 3.1 V should be expected during wake-up; the duration of these dips should not exceed tens of microseconds. Refer to [MM APPNOTE-34 - Measuring MM6108 Low Power]() for additional details.

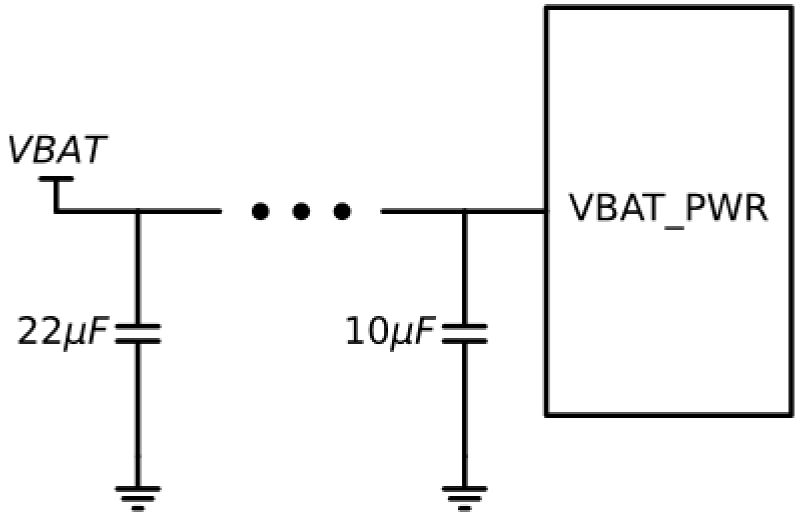


Figure 15. Bulk decoupling capacitors for the main MM6108 voltage supply VBAT_PWR

In applications where tighter tolerance is required on the main power supply, additional capacitance should be applied to VBAT_PWR. For example, if VBAT_PWR is 3.3 V and cannot fall below 3.25 V, at least 110 µF of decoupling capacitance should be provided. All decoupling capacitors should use an X7R dielectric or equivalent.

⚠️Note: the effective capacitance of ceramic capacitors can drop significantly under applied DC bias, especially in smaller packages. For example, a nominal 4.7 µF capacitor in 0402 package may deliver less than 2 µF at 3.3 V. When selecting capacitors for bulk storage, consult the manufacturer's bias derating curves to ensure that sufficient capacitance remains available under load. If necessary use a larger case size or a higher voltage rating to reduce the percentage loss.

4.7. Interfacing

4.7.1. SDIO / SPI Interface

The MM6108 defaults to SDIO mode after power-on. To use SPI instead, the host must issue a CMD0 command with the chip select (CS) line held low. This places the device into SPI mode for the remainder of the session. The EXT_HOST_SEL pin must be tied high to enable host communication via SDIO or SPI.

Pin	Name	SDIO 4-bit mode	SDIO 1-bit mode	SPI Mode
31	SDIO_CLK	Clock pin (input)		SPI_SCK

30	SDIO_CMD	Command pin		SPI_MOSI
29	SDIO_D0	Data pin 0		SPI_MISO
28	EXT_HOST_SEL	SDIO/SPI/QSPI interface enable strap (tie high)		
27	SDIO_D1	Data pin 1	IRQ	SPI_INT
26	SDIO_D2	Data pin 2	-	-
25	SDIO_D3	Data pin 3	-	SPI_CS

Table 4. Pinout for SPI and SDIO interfacing

The SDIO_CLK line must not include a pull-up resistor. The remaining SDIO lines (SDIO_CMD, SDIO_D0, SDIO_D1, SDIO_D2, and SDIO_D3) must be pulled up to VDDIO using resistors between 10 kΩ and 100 kΩ. The responsibility of providing these resistors falls on the host side; however, if the host does not provide them or its implementation is uncertain, it is acceptable to place the resistors on the peripheral (the MM6108) side to ensure correct logic levels.

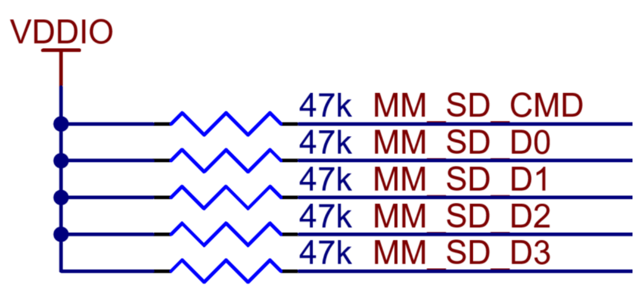


Figure 16. SDIO pull up resistors

SDIO traces should be routed with 50 Ω characteristic impedance. Basic length matching to within 10 mm is also recommended. If signal integrity issues are observed, such as excessive trace length or overshoot, series damping resistors in the range of 10 Ω to 100 Ω may be beneficial. Where signal behaviour is unknown at design time, footprint provisions for 0 Ω resistors can be included to preserve the option to correct signal integrity issues discovered during verification.

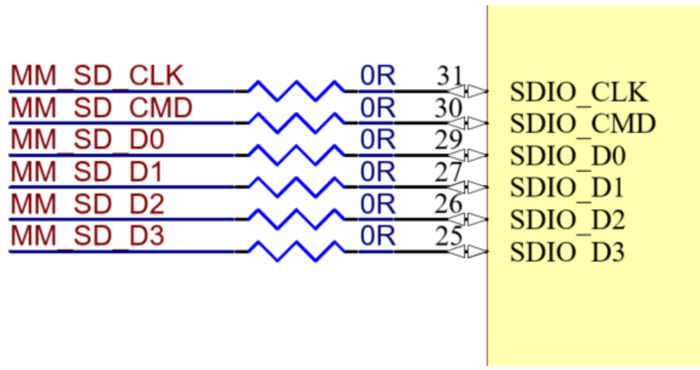


Figure 17. Placeholder series damping resistors for SDIO

When multiple host controllers are supported through different build options, it is critical to include explicit 0 Ω resistors (or similar jumpers) to disconnect unused signal paths from the SDIO lines. Simply leaving an unused host controller interface unpopulated is not enough; the signal traces should be physically isolated to prevent unterminated stubs that can cause signal reflections and degrade SDIO signal integrity.

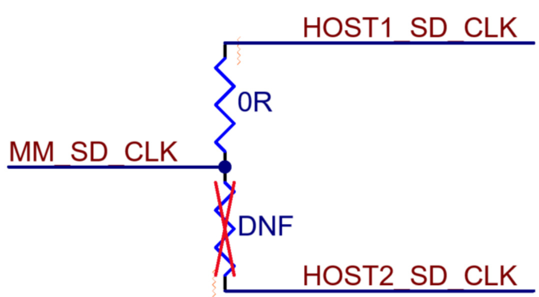


Figure 18. Selection of SDIO host controller via 0 Ω resistors

4.8. JTAG

The JTAG interface consists of five signals: TDI, TDO, TCK, TMS, and TRST.

Signal	Description
TDI	Test Data In
TDO	Test Data Out
TCK	Test Clock

TMS	Test Mode Select
TRST	Test Reset

Table 5. JTAG signals

The JTAG lines should be pulled to ground via 10 kΩ resistors.

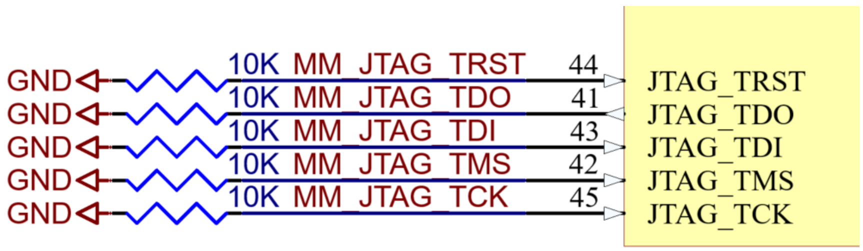


Figure 19. Pull-down resistors on the JTAG signal lines

Full impedance control is not required at these frequencies, but some effort should be made to avoid poor impedance mismatches. Basic length matching to within 10 mm is also recommended.

4.9. GPIO

GPIO may be used to control analog front end operation (e.g. LNA enable, PA enable, bypass switches, antenna diversity), indicator LEDs, or other digital functionality. Most GPIO signals operate at low speeds and do not typically require strict routing constraints for signal integrity. However, in applications where fast edge rates are present, attention should be given to trace length, return path continuity, and termination.

All GPIO lines should be terminated to ground using 10 kΩ pull-down resistors. During certain operating modes (such as reset or sleep) the GPIO pins may become undriven and enter a high-impedance state. Applying pull-down resistors ensures that these lines remain at a defined logic level, preventing noise-induced behavior such as spurious logic transitions or unexpected current draw.

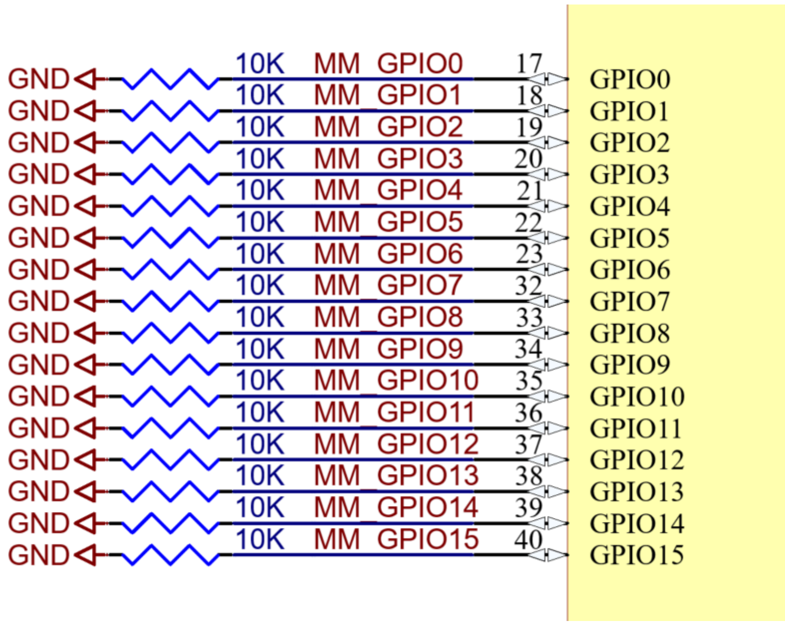


Figure 20. Pull-down resistors on unused GPIO lines

GPIO traces (especially those required to control the RF front end circuitry) can behave as resonant stubs, increasing susceptibility to coupling and radiated noise. For GPIO traces required to control the RF front end circuitry, placing a 100 pF shunt capacitor near to the downstream device should provide an effective short-circuit to ground for RF whilst applying minimal loading to the GPIO.

4.9.1. GPIO Pins with Special Functions

WAKE

Input signal to bring MM6108 out of deep sleep and snooze modes. This is part of the VBAT_PWR analog domain and cannot be driven by anything less than 3.0 V. Caution must be taken when the host controller operates using VDDIO less than 3.0 V to ensure appropriate level-shifting is employed when driving WAKE.

RESET_N

The active-low reset pin should be connected to a power-on reset network consisting of a 100 kΩ pull-up resistor and a 2.2 µF capacitor to ground. This ensures that the MM6108 is properly initialised upon power-up. The RESET_N pin can also be driven low by an external device to reset the MM6108.

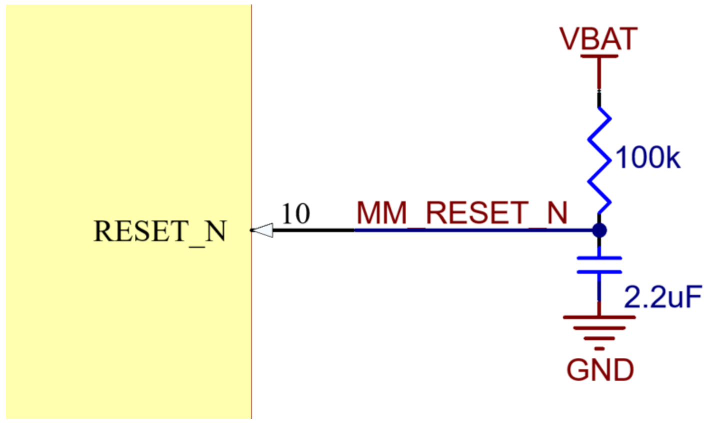


Figure 21. Power-on reset network

GPIO0

GPIO0 is configured by default as a digital output that signals when the MM6108 is busy.

GPIO10 - GPIO15

It is recommended to dedicate GPIO10 through GPIO15 to RF front end control functions such as PA, LNA, antenna switches, and related front end modules.

4.10. Diagnostics

4.10.1. Preferred Diagnostics

For robust design and enhanced troubleshooting capability, it is advisable to include a 16 pin header (Morse Micro uses Semtec FTSH-108-01-F-D-K) configured with the pinout below to allow access for potential diagnostics.

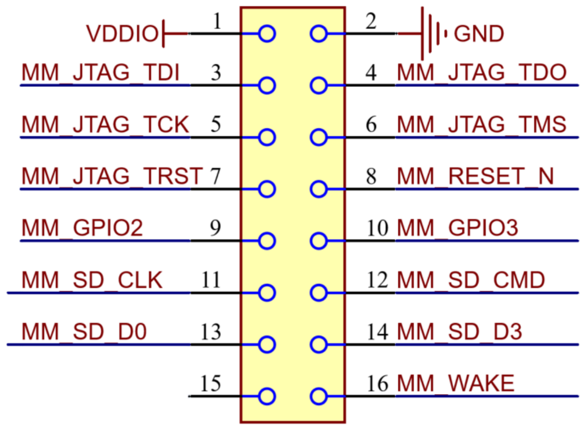


Figure 22. Standard pinout for diagnostics connector

The following pin configurations are recommended:

Pin	Net	Description
1	VDDIO	Digital domain power rail
2	GND	Ground
3	MM_JTAG_TDI	JTAG TDI
4	MM_JTAG_TDO	JTAG TDO
5	MM_JTAG_TCK	JTAG TCK
6	MM_JTAG_TMS	JTAG TMS
7	MM_JTAG_TRST	JTAG Reset
8	MM_RESET_N	Device Reset
9	MM_GPIO2	Debug
10	MM_GPIO3	Debug
11	MM_SD_CLK	SPI CLK
12	MM_SD_CMD	SPI MOSI
13	MM_SD_D0	SPI MISO
14	MM_SD_D3	SPI CS
15	-	Not connected

Pin	Net	Description
16	MM_WAKE	Wake up

Table 6. Standard pinout for diagnostics connector

4.10.2. Reduced Diagnostics

In designs where a full diagnostics header is not feasible, a reduced interface is supported. This consists of a 10POS 1.27mm pitch connector with the following connection:

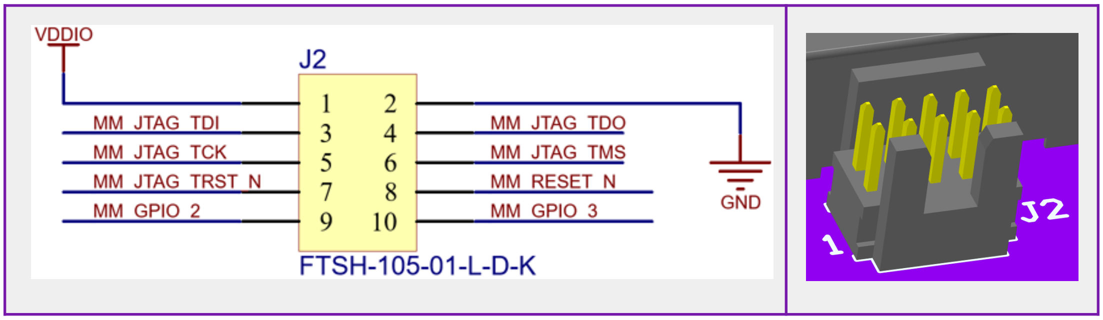


Figure 23. Pinout for alternative diagnostics connector

A diagnostics port is not required for normal device operation. However, if no debug access is provided in the design, the ability for Morse Micro to support, diagnose, or resolve technical issues will be reduced.

4.11. Protection

Standard best-practice electrostatic discharge (ESD) protection is recommended on all nets exposed to the external environment. This includes placing protection at the outermost connector, not at the MM6108. In designs that integrate the MM6108 into a module, ESD protection is not recommended within the module itself; it is the responsibility of the system that incorporates the module. Avoiding ESD components on the RF trace helps preserve signal integrity by minimizing parasitic shunting and impedance discontinuities. For the antenna trace, best results are achieved by selecting an antenna switch or front-end module with integrated ESD protection instead of applying additional discrete components. If external protection is unavoidable, the selected TVS diode must have an input capacitance less than 0.2 pF, and its footprint must preserve controlled-impedance geometry without introducing abrupt transitions.

Standard electromagnetic interference and electromagnetic compatibility design practices should be applied throughout. On the VBAT_PWR net, particular care must be taken to ensure that the effective series resistance (ESR) remains below 50 mΩ. Excess

ESR can lead to significant voltage droop during wake-up inrush events, potentially disrupting operation. If low ESR cannot be guaranteed, the power supply should support remote sensing or other compensation mechanisms to account for I^2R losses in the supply path.

The MM6108 is rated for 3.0–3.6 V. For nominal operation at 3.3 V, voltage clamping to 3.4–3.6 V is sufficient to protect the device from supply transients. The MM6108 does not include reverse polarity protection, so this must be implemented externally in systems where reverse polarity may occur (e.g. a user inserting a battery backwards). If adding voltage clamping or reverse polarity protection to a design, consider any trade-offs such as additional leakage currents. This is particularly important for low-power or battery-operated applications, where minimizing power consumption is critical.

5. PCB Layout

The following sections of this guide are organized to match the recommended workflow for a PCB layout engineer, addressing each area in turn; starting with placement—then routing—from the most critical elements (RF) down to the least (GPIO). By completing one section before moving to the next, this process naturally ensures that the most important parts receive the most favorable design trade-offs.

We start with some PCB layout examples before turning to the component placement and routing process in detail.

5.1. Radio Frequency

The RF section should be the primary layout consideration. It is recommended that it be routed first so that it is implicitly given the most favourable design trade-offs.

5.1.1. Placement Guidelines

It is advisable to start placing components with the antenna, followed by the front-end circuitry and then the HaLow transceiver module. This allows the RF section to be tightly packed with short, direct traces. During this process many design trade-offs will emerge. The following section details the key considerations to be mindful of when placing the RF components.

5.1.1.1. Path Length

Place RF components close to keep traces as short as possible. This will reduce insertion loss, improve transient settling, and minimize electromagnetic interference. Longer traces introduce greater resistive and dielectric loss, which reduces efficiency. A shorter trace also limits the round-trip path for reflections, allowing signals to settle more quickly after a transition. Shorter paths reduce the physical aperture available for radiated emissions or coupling to adjacent circuitry.

As a guideline, RF traces should be kept significantly shorter than one-tenth of the signal wavelength in the PCB dielectric. For typical coplanar waveguide or microstrip traces on FR4 at 915 MHz, this corresponds to approximately 15 mm. Traces longer than this threshold must be treated with increased attention to impedance control and layout. However, minimizing length is beneficial in all cases: Ohmic and dielectric losses scale directly with trace length, and even short traces will exhibit improved efficiency and reduced insertion loss when length is minimized.

5.1.1.2. Current Loops

Orient and place components that connect to ground (like decoupling capacitors or terminations) so their ground pads are as close as possible to vias or traces leading to the ground plane. Minimize the area enclosed by the signal and its return path, as this reduces loop inductance and magnetic coupling. Component placement should ensure that PCB traces are not routed beneath the RF front end circuitry.

5.1.1.3. Separation from Noisy Sources

Keep RF components and traces well separated from all other circuitry. As a general guideline, maintain at least 10 mm clearance from other components and traces, and maximize distance to switching regulators, high-speed digital lines, and potential electromagnetic interference sources. Layout choices should prioritize clean isolation, especially in sensitive receive paths or matched impedance networks, because even low-level coupling can degrade performance. It is advisable to define explicit keep-out zones around RF traces, matching networks and the RF front-end-module to prevent electromagnetic coupling and preserve circuit isolation.

5.1.1.4. Parasitic Impedances

When placing RF components, remember that PCB traces introduce parasitic inductance, while component pads add parasitic capacitance. Extra trace length is generally less critical for inductors, just as added copper area has less impact on capacitors. Knowing in advance whether a matching network will require inductors or capacitors (rather than relying on generic pi or tee footprints) allows placement and routing to minimize parasitic inductance at capacitors and parasitic capacitance at inductors.

5.1.1.5. Inductor Considerations

Careful placement and orientation of inductors is essential to minimize magnetic coupling, which can degrade RF performance or inject noise into sensitive circuits. When multiple inductors are used in proximity (such as RF matching inductors and DC-DC converter inductors) their coils should be placed with orthogonal orientations. Avoid placing inductors with parallel coil axes, as this maximizes coupling. Maintaining physical separation between inductors further reduces coupling risk.

These considerations are especially important during RF reception, when the system is most sensitive to low-level noise. In this state, any coupling between switching inductors and RF paths can compromise receiver performance. In particular, inductors inside matching circuits can act as antennas that pick up radiated noise.

5.1.1.6. Matching Component Considerations

Matching components should be placed as close to the unmatched element (e.g. the antenna) as possible. PCB traces before (or between) the matching components should not be viewed as conventional 50 Ω transmission lines, so keeping these traces to a minimum length reduces unwanted impedance transformation. Some advanced matching techniques may intentionally introduce phase shifts using this approach; however, such techniques are outside the scope of this guide.

5.1.1.7. PCB Antenna Considerations

Follow the antenna manufacturer's recommended keep-out dimensions carefully and measure antenna performance (including return loss and radiation pattern) as part of the design process. Keep in mind that PCB layout, nearby structures, and the enclosure can significantly influence antenna behavior and the specific interactions of your product are not captured in the antenna datasheet.

5.1.1.8. Shielding Considerations

A shielding can is strongly recommended over the entire RF section, including the HaLow chipset and RF matching networks. Switch-mode power supplies may require additional shielding cans. Keep in mind that inductors are particularly prone to radiate and receive noise, and during RF receive, even low-level coupled noise can degrade sensitivity or introduce spurious responses. The shield should fully enclose the sensitive region, with a solid via-stitched ground perimeter to ensure good RF sealing. Large openings or poorly grounded shields can significantly compromise effectiveness.

The primary purpose of shielding is to prevent emissions from other system components (particularly high-speed digital switching, or RF) from coupling into the antenna path and degrading receiver sensitivity. The device is capable of detecting as low as −107 dBm (and even lower if a low noise amplifier is employed), so even very weak internal emissions can raise the noise floor and limit system performance. Effective shielding should form a continuous seal. The limiting factor in shielding effectiveness is typically the largest dimension of any aperture or discontinuity in the shield—not the overall coverage percentage. Shielding with large openings often provides little benefit and can introduce resonances or coupling paths that degrade performance. At 915 MHz (λ ≈ 328 mm), the reactive near field extends approximately 50 mm from any source of interference. Within this region, electromagnetic energy can couple into the antenna even in the absence of a direct line-of-sight path (fields can diffract, wrap around shielding edges, or penetrate narrow slots).

5.1.2. Routing Guidelines

Once component placement is complete, begin routing the RF traces. Each RF trace can significantly impact system performance and EMC/EMI compliance, so route them with particular care and deliberate thought. The major considerations are detailed in the following section.

5.1.2.1. Impedance Control

RF traces should be designed for a 50 Ω characteristic impedance. The prior layout examples use coplanar waveguide traces (width approximately 0.2 mm, gap approximately 0.2 mm) on a four layer FR-4 PCB (where the substrate thickness from the signal layer to ground layer is approximately 0.1 mm). The exact width, gap and thickness required are dependent on the fabrication process and materials used. For this reason, it is essential that you coordinate directly with your PCB manufacturer during layout to define final trace dimensions using their specific stack-up and material properties.

5.1.2.2. Bend Angle

Smoothly curved or 45° bends in RF traces are preferred. Sharp corners can introduce minor discontinuities, so gradual transitions are recommended to support consistent impedance and clean signal propagation. A 45° bend or continuous curve provides a more uniform current path, reducing the chance of localized reflections or unintended radiation.

5.1.2.3. Reference Plane Integrity

Route RF traces exclusively over an uninterrupted ground plane; do not place RF traces over plane splits, gaps, other traces, or isolated copper pours. An unbroken ground plane provides a consistent return path directly beneath the RF trace, which is essential for stable impedance and minimising unintended radiation.

5.1.2.4. Via Usage and Transitions

RF paths should remain on the outermost copper layer and avoid layer transitions via vias. Routing RF traces entirely on the surface avoids introducing impedance discontinuities and parasitic via inductances. Layer changes inherently degrade signal integrity and are strongly discouraged in RF paths. If an unavoidable mechanical constraint forces a layer change, only then should a single, properly sized via be used, with minimal stub length (preferably back-drilled), and impedance-matched barrel and antipad dimensions.

5.1.2.5. Via Shielding

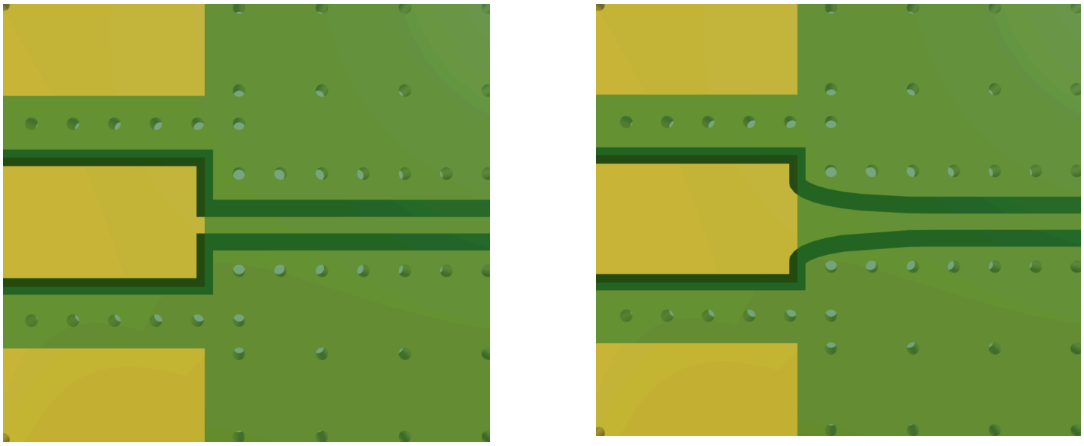
Coplanar waveguide (CPW) traces should be flanked on both sides by a continuous *fence* of closely spaced ground vias. This via fencing confines the electromagnetic fields, suppresses parallel-plate modes, and maintains a robust and continuous return path, which minimizes undesired coupling and radiation. Vias should be positioned as close to the edge of the CPW ground metal as practical; ideally within 0.5 to 1.0 mm. Denser via spacing improves mode suppression and helps ensure consistent impedance and field confinement along the CPW structure.

5.1.2.6. Thermal Relief and Power-Plane Intersections

For optimal signal integrity, ground vias associated with RF signals should use a solid copper pour rather than thermal relief patterns. This ensures the lowest possible return path impedance and minimizes parasitic effects. However, solid vias can conduct significant heat during soldering, potentially causing assembly challenges or unreliable joints. Thermal reliefs alleviate this risk by limiting heat conduction—but at the expense of RF performance. We recommend retaining a solid ground connection and addressing assembly concerns through process tuning or other local design adjustments.

5.1.2.7. Component Pad Transitions

When transitioning to a wider pad, an abrupt and well-controlled impedance transition is generally preferable, particularly when coordinated with an appropriate ground return (such as removing some of the ground layers under the pad, if needed). Tapered transitions may look cleaner but often do not improve impedance matching and can worsen reflections.



Without taper - preferable

With Taper - not preferable

Figure 24. Transition from 0.2mm trace to SMA connector

In the example shown in Figure 24, the RF feedline is designed for a 50 **Ω** characteristic impedance, referenced to an internal ground plane. However, the RF connector pad is significantly wider than the feedline. To maintain the 50 **Ω** impedance at this interface, the internal ground layers are removed beneath the pad, shifting the reference plane to the bottom layer. On the left, an abrupt transition is shown where the trace width and ground reference shift simultaneously—this preserves a consistent 50 **Ω** impedance across the transition. On the right, a tapered trace is used instead. As the trace widens while still referenced to the internal plane, the impedance gradually drops below 50 **Ω** before returning once the bottom-layer ground plane takes over at the pad. This non-uniform impedance can introduce reflections and degrade RF performance.

The best results are achieved by avoiding unnecessary trace width discontinuities altogether. Selecting components (e.g. 0402 passives) and connectors whose pad dimensions are as close as possible to the desired trace width is recommended. However, the trace width is also typically constrained by the need to mate with the device's interface geometry (e.g. the required pad or bump size), which may impose a minimum trace thickness. In practice, the thinnest required pad length often determines the trace width, which in turn defines the PCB stackup and transmission line type (e.g. CPW or microstrip).

Transitions to larger connector geometries (such as SMA or U.FL) can introduce impedance discontinuities due to differences in pad size compared to the RF trace width. These transitions should ideally be measured using a vector network analyzer to verify return loss and ensure acceptable matching. While full 3D electromagnetic simulation is not mandatory, it can be a valuable tool for predicting performance and reducing the number of layout iterations needed to optimize RF performance.

5.2. Buck Converter

The buck converter produces high-frequency, high-current signals that flow through an inductor and the ground plane. Careful design is needed to prevent these from coupling or radiating into the receiver, crystal, or bypass capacitors, which could degrade HaLow performance. It is advisable to place and route this section only after the RF circuitry is in a stable layout state.

5.2.1. Placement Guidelines

The figure below illustrates a typical layout. The inductor, bulk capacitor, and bypass capacitor (highlighted in Figure 27) should all be placed on the same side of the PCB as the MM6108 to minimize via transitions, tighten current loops, and maintain clean return paths.

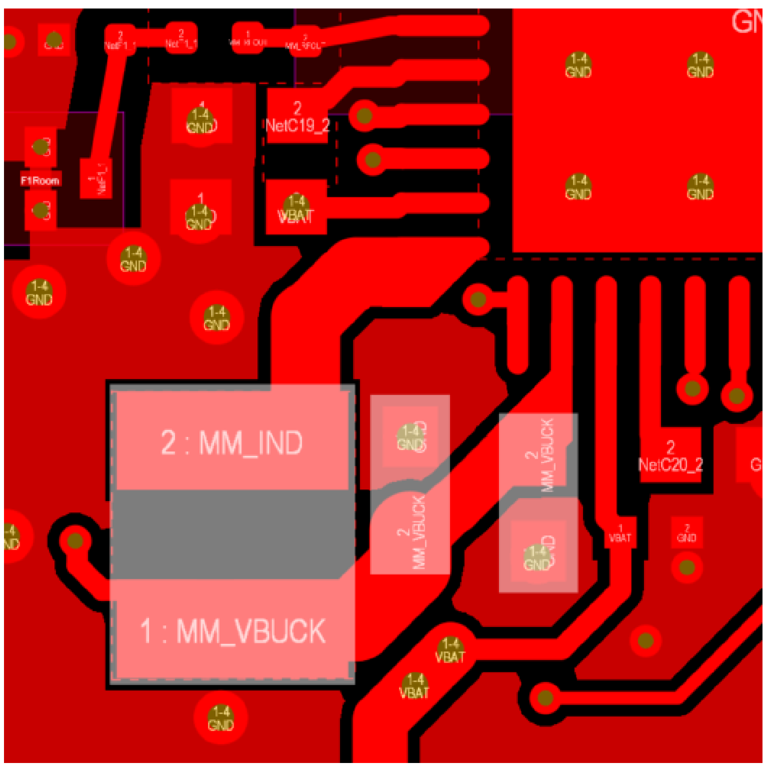


Figure 27. Buck converter component placement

5.2.1.1. Inductor

Place the inductor directly between the IND pin and the shared output node that serves the capacitors and the VBUCK pin. It must be located as close to the IND pin as practical. Even shielded inductors leak some magnetic field; to reduce coupling into nearby signals, keep sensitive traces away from the inductor's sides. If placement near another inductor is unavoidable, ensure the inductors are rotated by 90 degrees relative to each other to minimize mutual coupling. It is also advisable to evaluate whether eddy currents in the ground plane (induced by the coil's magnetic field) could be an issue; if so, a ground pour cutout beneath the inductor may help mitigate them.

5.2.1.2. Bulk Capacitor

Place the 10 µF capacitor immediately at the shared node where the inductor output connects. Its positive terminal should be as close as possible to the inductor's output pad. Its ground pad should connect directly back to the ground plane using at least one via placed immediately at the pad, minimizing impedance and loop area.

5.2.1.3. Bypass Capacitor

Place the 4.7 µF capacitor immediately at the VBUCK pin. Its positive terminal should be as close as possible to the IC. Its ground pad should connect directly back to the IC ground using at least one via placed immediately at the pad to keep the return path short and low impedance.

5.2.2. Routing Guidelines

The routing process sets the parasitic impedances and determines the area of current loops, all of which should be kept as small as possible. Route with careful attention to the specific current loops present in a buck converter, as detailed in the following section.

5.2.2.1. Switching Loop (Hot Loop)

This loop carries high-frequency switching current and includes the path from IND, through the external inductor, into the output capacitor, and back to the IC via the capacitor's ground connection. It is a potential source of electromagnetic interference in the system and must be routed with great care. The trace from the inductor to the capacitor must be short and direct, and the capacitor's ground pad must connect to the return path using at least one via placed directly at the pad. This loop should be tight and isolated from unrelated signals and copper to minimise both inductive and capacitive coupling.

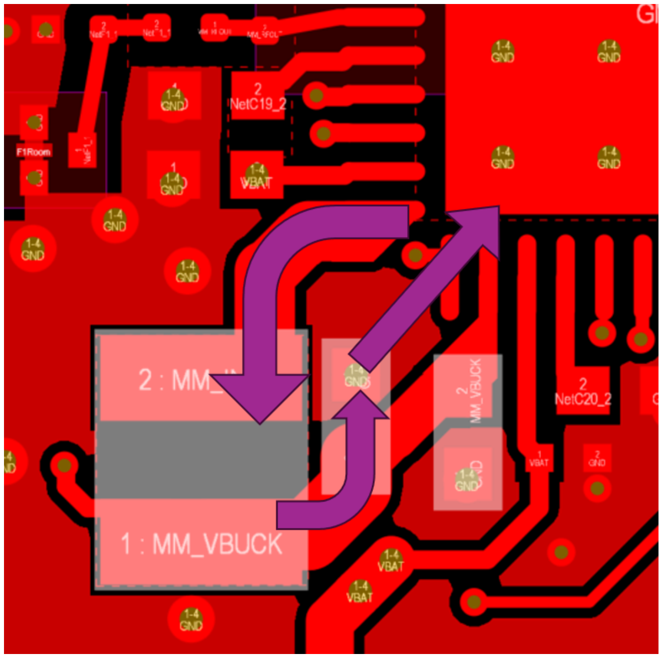


Figure 28. Buck converter hot loop current path

5.2.2.2. Direct Current Path

The direct path carries load current from IND, through the inductor, into the VBUCK pin, which supplies the internal load. The inductor output also connects to the output capacitors, so the routing must not compromise these connections. The VBUCK trace should tee from the shared output node at the inductor–capacitor junction, but only after the bulk capacitor connection is established. The VBUCK trace must also be short and wide and should not interfere with the capacitor’s placement or return path.

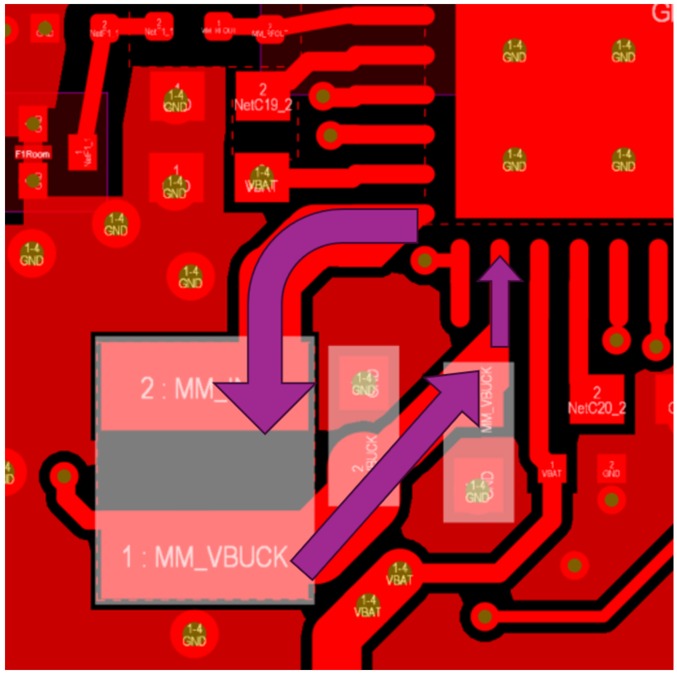


Figure 29. Buck converter direct loop current path

5.2.2.3. Capacitor Discharge Loop

This loop conducts when the output capacitors supply current to the load. It shares the same termination points as the direct path (i.e. VBUCK); if trade-offs arise the bypass capacitor connection should take priority because it sets the output voltage. The ground return path should not be shared with unrelated current paths. Keeping this loop compact and low in impedance helps maintain voltage stability during switching transitions.

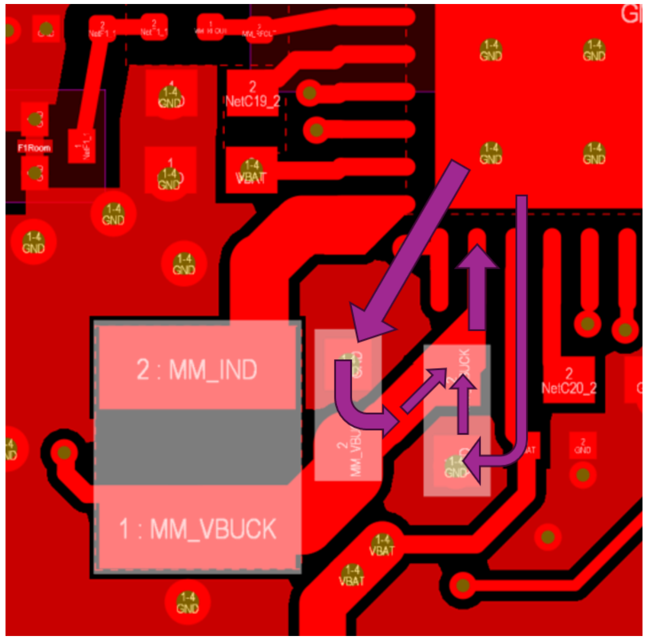


Figure 30. Buck converter capacitor discharge loop current path

5.2.2.4. Feedback Voltage Sense

The VBUCK_FB pin provides the voltage feedback to the internal regulator. It must be routed as a Kelvin connection from the inductor, avoiding any shared or high-current paths. The trace should not pass through (or beneath) the inductor or any load connection. If a via is required, ensure that it does not share other return currents. This ensures accurate voltage sensing and stable regulation.

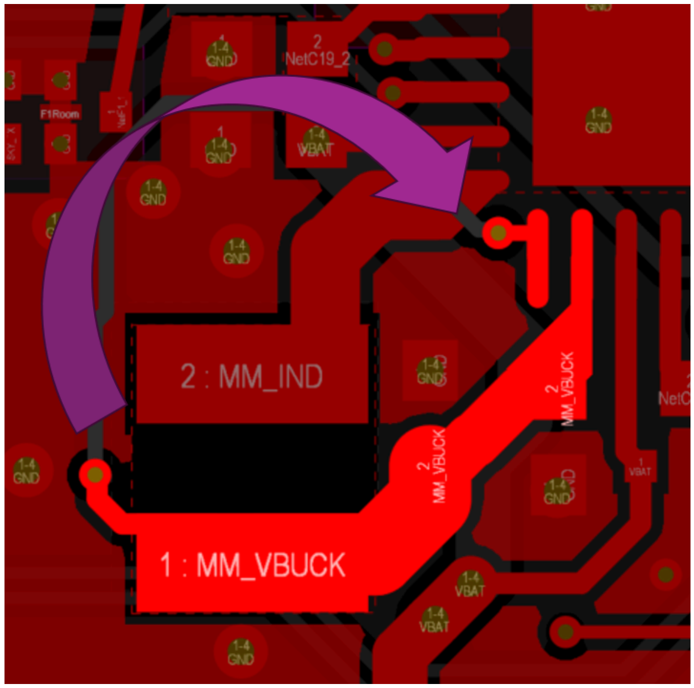


Figure 31. Buck converter Kelvin connection for voltage sense feedback

5.2.2.5. Grounding

The ground return for the VBUCK output capacitor must be made directly into a solid, continuous ground plane. Do not route return current through split or discontinuous planes, which would increase the impedance and loop area. Minimizing the area of the switching and output current loops is critical for reducing electromagnetic interference and ensuring efficient, low-noise operation of the buck converter.

5.3. Decoupling

After the buck converter, the next most important current loops to minimize are the bypass capacitors. Standard best-practice principles apply, as discussed in the following section. The figure below illustrates a typical layout.

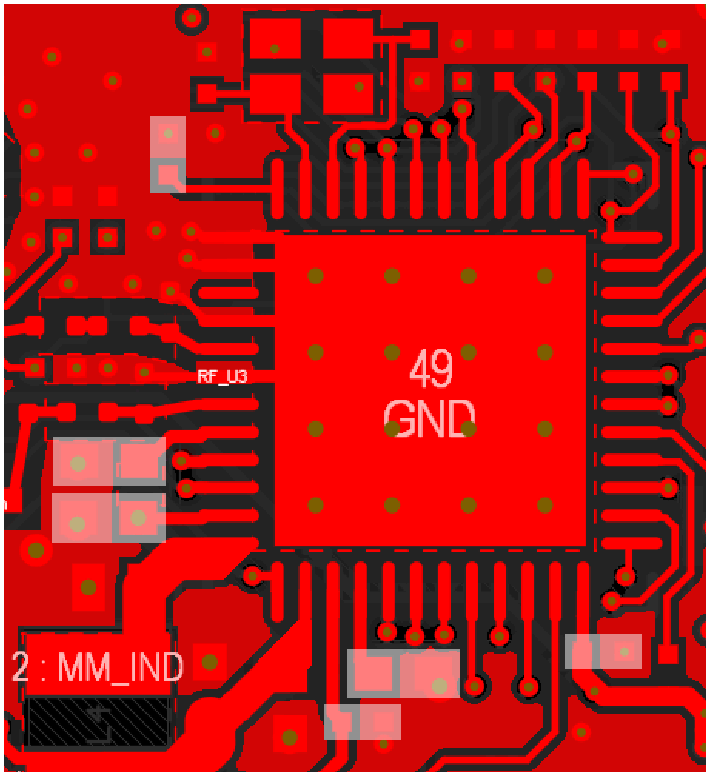


Figure 32. Decoupling capacitor placement

5.3.1. Placement Guidelines

Decoupling capacitors are most effective when placed to minimize loop area between the capacitor and its associated supply pin. All capacitors should be located as close as practical to their associated pins or supply nodes. When multiple capacitors are placed in parallel at the same pin (for example a 100 nF alongside a 4.7 µF) it may be acceptable to place them side by side or to stack them vertically (i.e. on opposite sides of the PCB with a via connection). What matters most is minimizing the inductance of the conductance path and ensuring both capacitors return to a solid ground plane. For designs with long or resistive supply paths on VBAT_PWR, placing sufficient bulk capacitance close to the device is especially important to ensure that transient current demands are supported locally.

5.3.2. Routing Guidelines

Use short, wide copper traces to both power and ground. A ground via should be placed as close as practical to the capacitor ground terminal to connect directly to a continuous ground plane located on the layer below the chip. This ground plane must be unbroken within the area enclosed by the outermost grounding vias of the chip and its decoupling capacitors, to provide a well-defined low-impedance current return path.

5.4. Power Supply

With the decoupling capacitors applied, it is now time to turn attention to the rest of the power supply architecture.

5.4.1. Placement Guidelines

Placement should aim to simultaneously minimize power supply noise and series resistance. Any noise on the power supplies may be coupled into the RF section of the design and then radiated out of the antenna. This can impact receive sensitivity and also cause the device to fail certification testing.

5.4.1.1. Equivalent Series Resistance

Equivalent series resistance (ESR) may be reduced by placing the power supply—particularly if it is a low dropout regulator (LDO)—close to the VBAT_PWR pin.

5.4.1.2. Separating Noisy Sources from Sensitive Circuitry

If using a switch-mode power supply, place it as far from the HaLow IC as practical. This applies broadly to any circuitry generating high-frequency currents through inductors or ground paths. If ESR in the main supply path is a concern, providing a Kelvin connection to the voltage feedback pin can help mitigate voltage drops.

5.4.2. Routing Guidelines

The routing of power supply traces directly influences voltage stability, noise coupling, and how effectively decoupling performs under dynamic load. The major considerations are detailed in the following section.

5.4.2.1. Star Topology Distribution

Keep all common mode impedance to a minimum. Power distribution should avoid daisy-chaining and instead use a star-point or short branch to deliver equal-quality supply to each load. This is required for VBAT_PWR and VBAT_AON, which serve distinct functional domains and must be routed independently from a shared node. Each branch must include its own local decoupling network placed close to the associated power pin. The physical layout should favor short, direct traces from the star point to each load. Route traces to the capacitor pads first and then to the device pin.

5.4.2.2. Trace Width and Thickness

Power supply traces should be designed for both low resistance and low inductive impedance to support fast current transients without significant voltage droop. The maximum recommended series resistance on the VBAT_PWR current path is 50 mΩ. The PCB layout should ensure that this limit is not exceeded, taking into account the resistance of the copper traces as well as any series components present in the power path. Using wider copper traces (or better yet, polygon pours) minimizes voltage drop due to DC resistance and reduces inductive impedance at higher frequencies. This becomes especially important during events such as RF transmission or wake-from-sleep transitions, where dynamic current demand can rise sharply. Supply voltage drop between the regulator and the load should remain below 50 mV under peak current conditions. Avoid routing supply traces through narrow necks, unnecessary detours, or shared segments that may introduce excess impedance or coupling into other supply domains.

Avoid overlapping power traces on adjacent layers, which can create capacitive coupling between supplies.

5.4.2.3. Via Usage

Each via contributes its own resistance and inductance, so spreading current across several vias ensures that power integrity is maintained even under fast load steps. This is particularly important for VBAT_PWR and the power supply to any RF power amplifiers.

5.5. Reference Crystal

The crystal oscillator sets the fundamental timing for the system, making it one of the most sensitive nodes to noise and layout imperfections. Even small amounts of coupled interference or imbalance in the layout can degrade phase noise or introduce spurs that impact overall RF performance. An example layout is provided below.

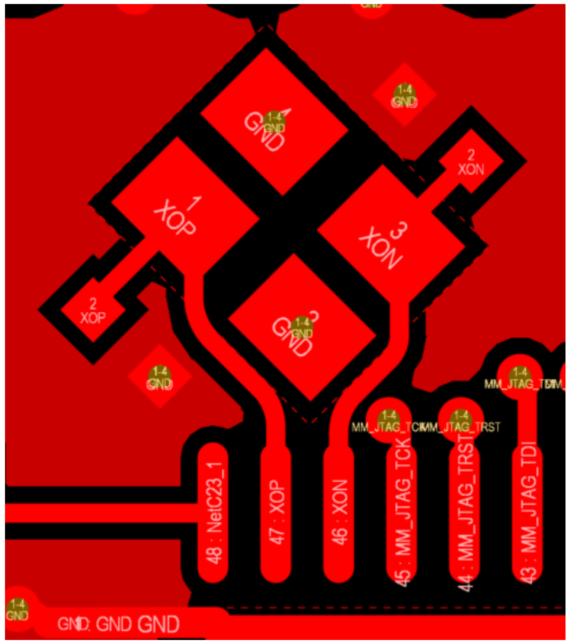


Figure 33. Reference crystal placement and routing

5.5.1. Placement Guidelines

To minimize noise coupling and maintain signal integrity, the crystal and its associated capacitors must be placed close to the oscillator pins, with the shortest possible traces between them. As a rule of thumb, no such signals should be routed within 10 mm of the crystal circuit.

5.5.2. Routing Guidelines

For best results, the crystal nets XOP and XON should be routed as a differential pair with matched lengths, keeping them short unless there is good reason for additional length (e.g. improving temperature stability by keeping the crystal away). A dedicated local ground pour should be provided underneath the entire crystal network, and signal traces (especially switching or high-speed digital lines) must not be routed underneath or near the crystal. Avoid routing stubs or introducing T-junctions in the connections to the oscillator, and ensure symmetrical routing between the two sides of the crystal to maintain balanced loading.

5.6. SDIO / SPI / JTAG

Placement and routing should minimize crosstalk and signal integrity issues across these digital interfaces. It is important to address this carefully during layout, as digital signals may appear functional despite poor integrity, often leading to intermittent or hard-to-diagnose failures later in development or in the field. The following recommendations help maintain clean digital signaling.

5.6.1. Placement Guidelines

Wherever possible, arrange subsystems on the PCB to allow direct, uncluttered routing of the SDIO, SPI, and JTAG signals, minimizing crossings with other traces or with each other. Maintain separation between these signal domains to reduce coupling and simplify layout.

5.6.2. Routing Guidelines

The SDIO and SPI signals should ideally be routed as a 50 Ω coplanar waveguide (on outer layers) or as stripline (on internal layers). To preserve signal integrity, keep trace lengths under 50 mm and match them within ±10 mm. Being digital signals, resistive and dielectric losses are not a key concern, so minimum trace width and dielectric type are not critical.

These interfaces run at tens of megahertz, so tight impedance control isn't necessary; however, using a standard PCB calculator to size traces for nominal 50 Ω is still recommended. Coordination with the PCB fabricator for precise impedance control on these traces is not mandatory.

SDIO is a point-to-point protocol and should always be routed directly between the host and peripheral without any branches in the signal path. When supporting multiple host devices through different builds (for example, by selectively populating 0 Ω resistors), the stub leading to the unused path should be kept as short as possible to maintain signal integrity. For shared buses like SPI or JTAG, arranging devices in a daisy chain helps minimize reflections and ensures the last device sees the best signal. Note that intermediate devices inevitably see stubs from the onward connections.

5.7. GPIO

GPIO signals have lower priority than other types, so they are typically routed last, using whatever space remains after higher-priority traces are placed. Ensure their routing doesn't violate existing layout constraints — for example, avoid cutting ground planes beneath RF traces or disrupting return current loops. GPIOs are *general purpose*; their function is defined by special firmware builds and Morse Micro *board configuration files* (BCF). If tied to a specific application note, any unique signal requirements will be detailed there.

Electrically, a GPIO driven on one end is effectively grounded at the driver but floating (high impedance) at the load. From the standpoint of nearby RF traces, it resembles a length of unterminated copper that can easily pick up or reradiate signals. If the trace is long or closely spaced to RF, it can act like a stray antenna. Adding 100 pF capacitors to ground can suppress these effects without impacting low-speed digital performance.

For GPIO signals that control external RF front-end devices such as RF switches, front-end modules (FEMs), low noise amplifiers (LNAs) and power amplifiers (PAs), ensure to place 100pF termination capacitors at the RF front-end device control input pins. These capacitors filter digital power supply domain noise and prevent this noise from coupling into the RF front-end signal path.

If GPIO traces need to cross on adjacent layers, or cross over power traces, ensure they do so at right angles to minimize coupling. GPIO traces should only run directly alongside other traces if those signals are tolerant of digital noise, or if GPIO transitions are guaranteed to occur at times that coupling is not consequential.

5.8. Diagnostics Ports

Placement and routing of the diagnostics connector should ensure reliable mechanical fit, allow unobstructed cable access, and maintain a clean integration into the PCB stack-up without compromising nearby traces or ground integrity. Place the diagnostics connector with enough clearance for cables or probes to attach and detach freely, and avoid positions that force sharp bends.

5.9. Protection

Placement and routing of protection components must ensure effective suppression of electrostatic discharge, electromagnetic interference, and conducted noise without compromising impedance control or power delivery.

5.9.1. Placement Guidelines

Protection components must be placed to intercept disturbances before they propagate into sensitive areas of the design. Electrostatic discharge protection circuitry should be located at the outermost interface with the external environment, close to connectors or antenna feeds, ensuring transients are shunted quickly through short, direct paths to ground. As signals transition between noisy and quiet domains, chokes and filters confine conducted noise, preserving the integrity of downstream circuits. On impedance-sensitive traces, particularly RF paths, any protection must be positioned to minimize parasitic effects and maintain controlled impedance without introducing discontinuities. Similarly, on power nets, protection layout must accommodate wide copper traces to keep ESR low, preventing voltage drops and avoiding inductive artifacts that could degrade supply stability.

5.9.2. Routing Guidelines

Route to protection devices with the shortest possible traces to minimize series impedance. Avoid tight corners or via transitions that introduce additional parasitics, particularly on RF or high-current power nets. Ensure ground returns are direct and connect to a solid plane to support effective discharge and filtering.

5.10. Grounding

Ground layout directly impacts power integrity, electromagnetic interference performance, and RF behavior. The following section describes layout techniques and grounding practices necessary to ensure stable operation. Many of the recommendations are repeated, but they have been kept together here so that this section may act as a final checklist on grounding integrity before the board is sent for manufacture.

5.10.1. Solid Ground Plane

Dedicate at least one layer (typically an inner layer) as a continuous ground plane. Avoid cutting or slotting the ground plane near critical components or between supply and decoupling points. Keep the ground plane unbroken under the MM6108, its associated circuitry and the RF front end. If signals must transition between layers, note that several adjacent vias will cause a split in the ground plane and must be spaced apart to prevent this. A single, unified ground plane should be maintained throughout. Do not segment or isolate analog and digital grounds within the IC footprint.

5.10.2. Ground Via Stitching

Use generous ground via stitching to connect all ground regions and layers. Every decoupling capacitor should have at least one ground via placed directly at the capacitor pad. Where possible, via-in-pad is preferred. Ground vias should be placed densely around exposed thermal pads, the periphery of the chip, and beneath any high-current return paths such as the buck regulator or RF output. Add ground vias at every ground pad, for every grounded component. For RF nets, use a direct connection (not a thermal relief).

In systems operating in the 850 MHz to 950 MHz band, the wavelength in FR4 is approximately 150 mm. To effectively confine return currents and suppress cavity resonances, stitching vias should be placed at least every 15 mm (about $\lambda/10$). However, for improved margin and to further minimize electromagnetic interference risk, a tighter pitch of 6 mm or less (approximately $\lambda/25$) is recommended.

Along critical signals (such as around RF or high-speed traces) vias should be spaced no more than 1.0 mm apart (center to center). Recommended via dimensions include a minimum drill size of 0.3 mm and a pad diameter of at least 0.6 mm, using annular rings that meet fabrication requirements. Place stitching vias every 3.0 mm along plane boundaries and within 1.0 mm of any cutout or slot.

5.10.3. Return-Path Control

All high-speed or RF signal traces must have a direct return path on the adjacent ground layer. Return current should remain closely coupled beneath the signal trace. Traces must not cross plane splits.

6. Application Example

This example is based upon the MM6108-MF08651-US reference design, which is an example application built around the Skyworks SKY66423-11 front end module (FEM). It comprises a power amplifier, low noise amplifier and antenna switch.

6.1. Bill of Materials

The following table lists the components required.

Designator	Value	Manufacturer	Part Number	Quantity
U1		Skyworks Solutions	* SKY66423-11	1
F1		Qualcomm	* B39921B2625P810	1
L4	10 uH	Murata	* DFE252012F-100M=P2	1
L8	5.6 nH	Murata	LQP03HQ5N6H02D	1
L9	8.2 nH	Murata	* LQP03HQ8N2H02D	1
L10	12 nH	Murata	LQP03HQ12NH02D	1
L1	6.8 nH	Murata	* LQW15AN6N8J00D	1
L2	1.2 nH	Murata	* LQP03HQ1N2B02D	1
C16, C19, C20, C22	4.7 uF	Yageo	CC0402KRX5R5BB475	4
C7, C15	10 uF	Samsung	CL05A106MP8NUB8	2
C6, C8, C23	470 nF	TDK	C0603X5R0J474M030B C	3
R3, R11, R12, R14, R15, R16, R19, R20	10 kΩ	Panasonic	ERJ-1GNF1002C	8
R5	62 Ω	Panasonic	ERJ-1GNF62R0C	1
C29	0.5 pF	Yageo	CC0201BRNPO9BNR50	1
C34	470 pF	Murata	GRM0335C1E471GA01 D	1
R1	0 Ω	Panasonic	ERJ-1GN0R00C	1
C28	15 pF	Yageo	CC0201JRNPO8BN150	1
C2	1 nF	Yageo	CC0201KRX7R7BB102	1
C1	10 nF	Murata	GRM033R61E103KA12 D	1
C4	33 pF	Murata	GRM0335C1H330JA01	1

Designator	Value	Manufacturer	Part Number	Quantity
			D	
C3, C11, C12, C13, C17, C18, C25, C31, C32	100 pF	Walsin Technologies	0201N101G500CT	9
C5, C24	12 pF	Johanson Technology Inc.	250R05L120JV4T	2
C9	3 pF	Johanson Technology Inc.	250R05L3R0BV4T	1
C14	2.4pF	Johanson Technology Inc.	250R05L2R4BV4T	1
C21	1.5pF	Johanson Technology Inc.	250R05L1R5BV4T	1
L3	2.7 nH	Johanson Technology Inc.	L-07C2N7SV6T	1
L5	7. 5 nH	Johanson Technology Inc.	L-07C7N5JV6T	1
R6, R7	100 Ω	Stackpole Electronics Inc	RMCF0201FT100RTR-ND	2
U2, U4		Richwave	* RTC6608OU	2
U3		Morse Micro	* MM6108	1
X1	32MHz	SiWard	* XTL901-M202-001	1

Table 11. Bill of materials for Antenna Switch + Low Noise Amplifier + Power Amplifier example (MM6108-MF08651-US Reference Design)

* These parts may not be substituted for equivalent components of the same value.

6.2. Reference Schematic

The Skyworks SKY66423-11 FEM provides an integrated RF front end, including a power amplifier (PA), low-noise amplifier (LNA), RF switch, and internal DC blocking. LNA bypass functionality is implemented using additional external RF switches (U2 & U3), as shown in the circuit below. A B39921B2625P810 SAW filter (F1) is included to improve EMC performance.

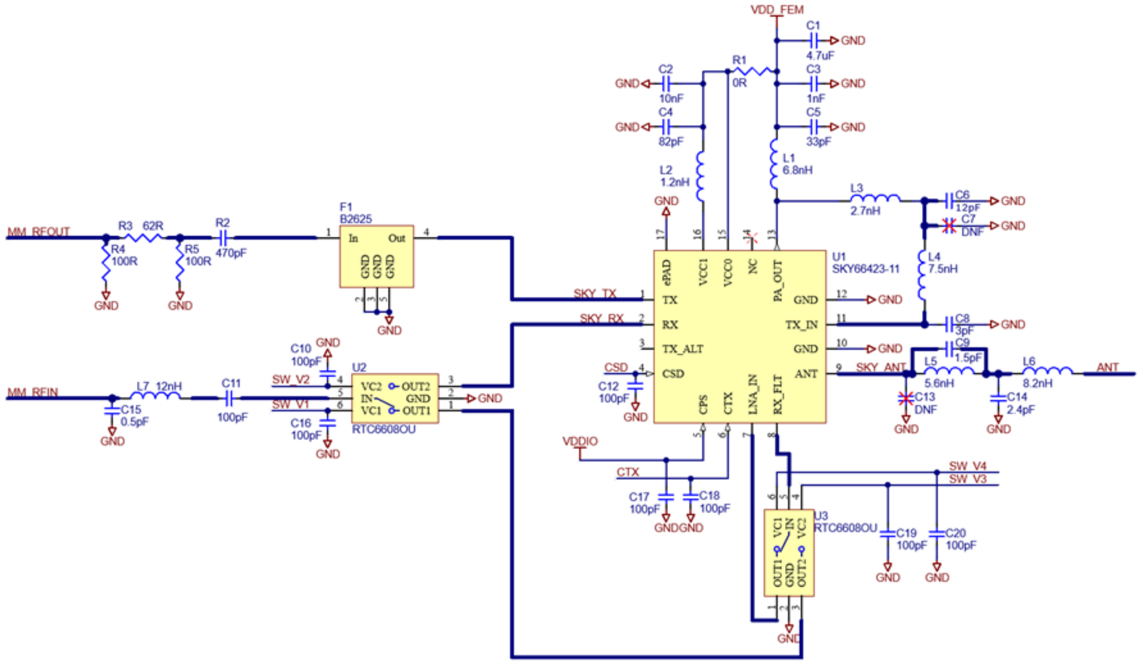


Figure 36. RF front end circuitry for Antenna Switch + Low Noise Amplifier + Power Amplifier example (MM6108-MF08651-US Reference Design)

The following GPIO configurations are recommended:

MM6108	Net	Description
GPIO10	SW_V4	LNA bypass switch control line
GPIO11	SW_V3	LNA bypass switch control line
GPIO12	SW_V2	LNA bypass switch control line
GPIO13	SW_V1	LNA bypass switch control line
GPIO14	CTX	FEM transmit mode select
GPIO15	CSD	FEM shutdown control

Table 12. MM6108 GPIO configuration for Antenna Switch + Low Noise Amplifier + Power Amplifier example

Approximate component values have been provided for the harmonic filter comprising inductors L5-6 and capacitors C9, C13-14; actual values will depend on the interaction between the components and parasitic impedances introduced by the PCB layout. This harmonic filter should be matched to 50 Ω, with minimal insertion loss and linear phase at the fundamental (850 MHz - 950 MHz) and at least 20 dB attenuation at the first harmonic.

Resistors R3, R4, and R5 form an attenuator with a 50 Ω characteristic impedance. For the above reference design, this provides approximately 9.2 dB of attenuation. The MM6108 has its best linearity when the transmit power presented at its output is approximately 0 dBm. Hence, for optimal performance, the attenuation provided by R3, R4 and R5 should be selected so that:

$$0\,dBm - attenuation + pa\ gain = desired\ output\ power$$

The impedance seen at U1 pin 16 (VCC1) and pin 13 (PA_OUT) should present a return loss of 0 dB at the fundamental frequency (850 MHz to 950 MHz). If this condition is not met, resonances caused by interaction with parasitic impedances in the decoupling circuitry may be present. In such cases, the values of C4 and C5 may be adjusted to shift these resonances out of the operating band. Refer to MM APPNOTE-29 - MM6108-MF08651 Bypass Boobytrap for additional details.

6.3. Reference Layout

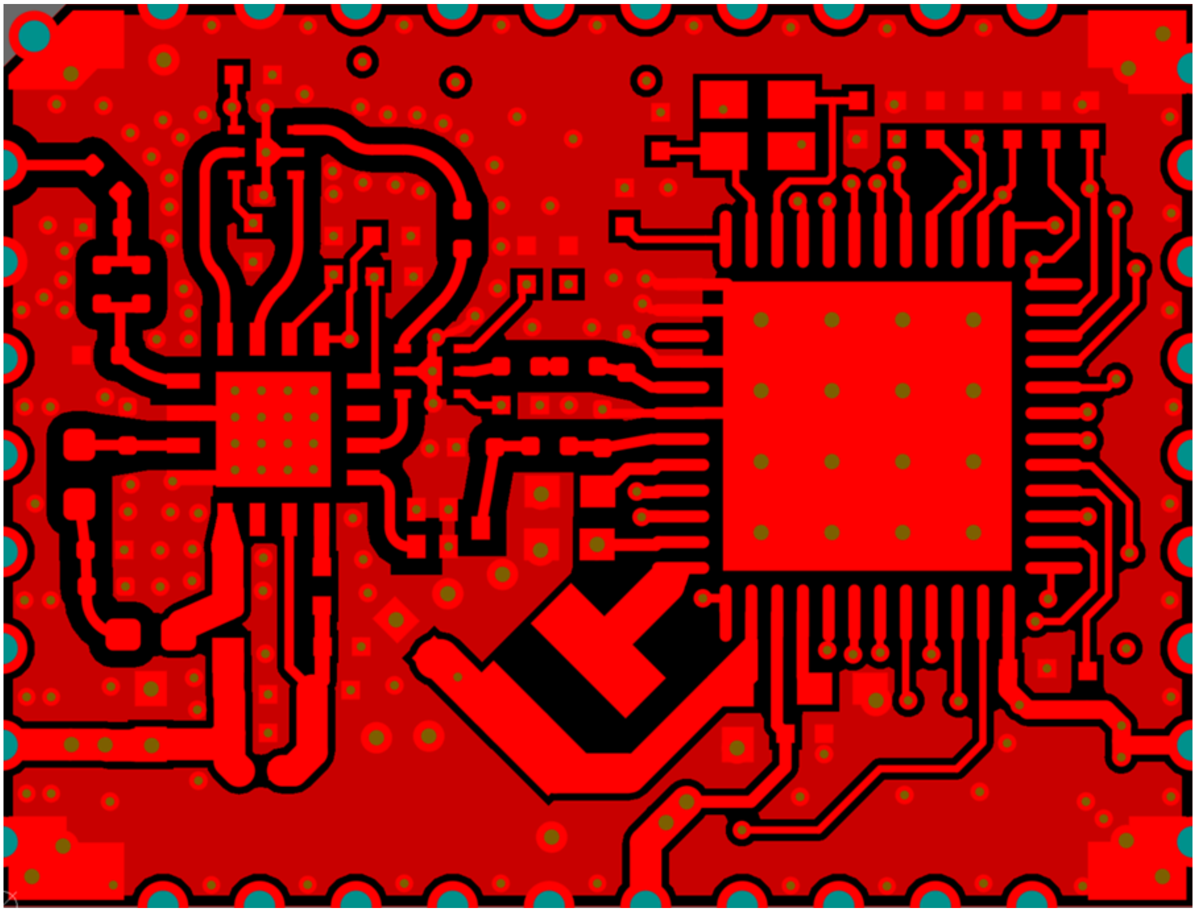


Figure 37. Layout for Antenna Switch + Low Noise Amplifier + Power Amplifier example (MM6108-MF08651-US Reference Design)

When laying out the power supply for the Skywork FEM, pay close attention to the routing to make it as close as possible to the Skywork recommended layout. This includes the first stage connecting to the VCC1 (16) and VCC0 (15) pins of the main PA supply going to PA_OUT (13).

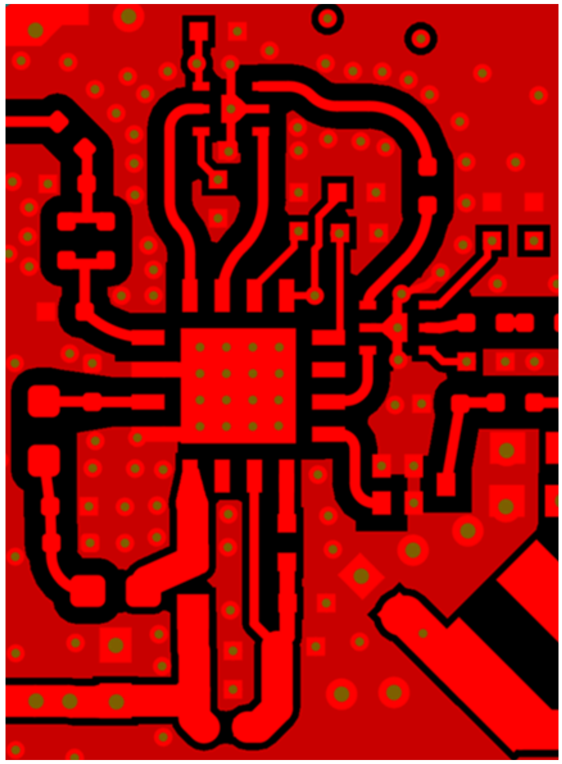


Figure 38. Layout for Skyworks front end module

7. Revision History

Release Number	Release Date	Release Notes
01	27/04/2022	Initial release
02	24/06/2025	Revised content, added sections, and updated design details.

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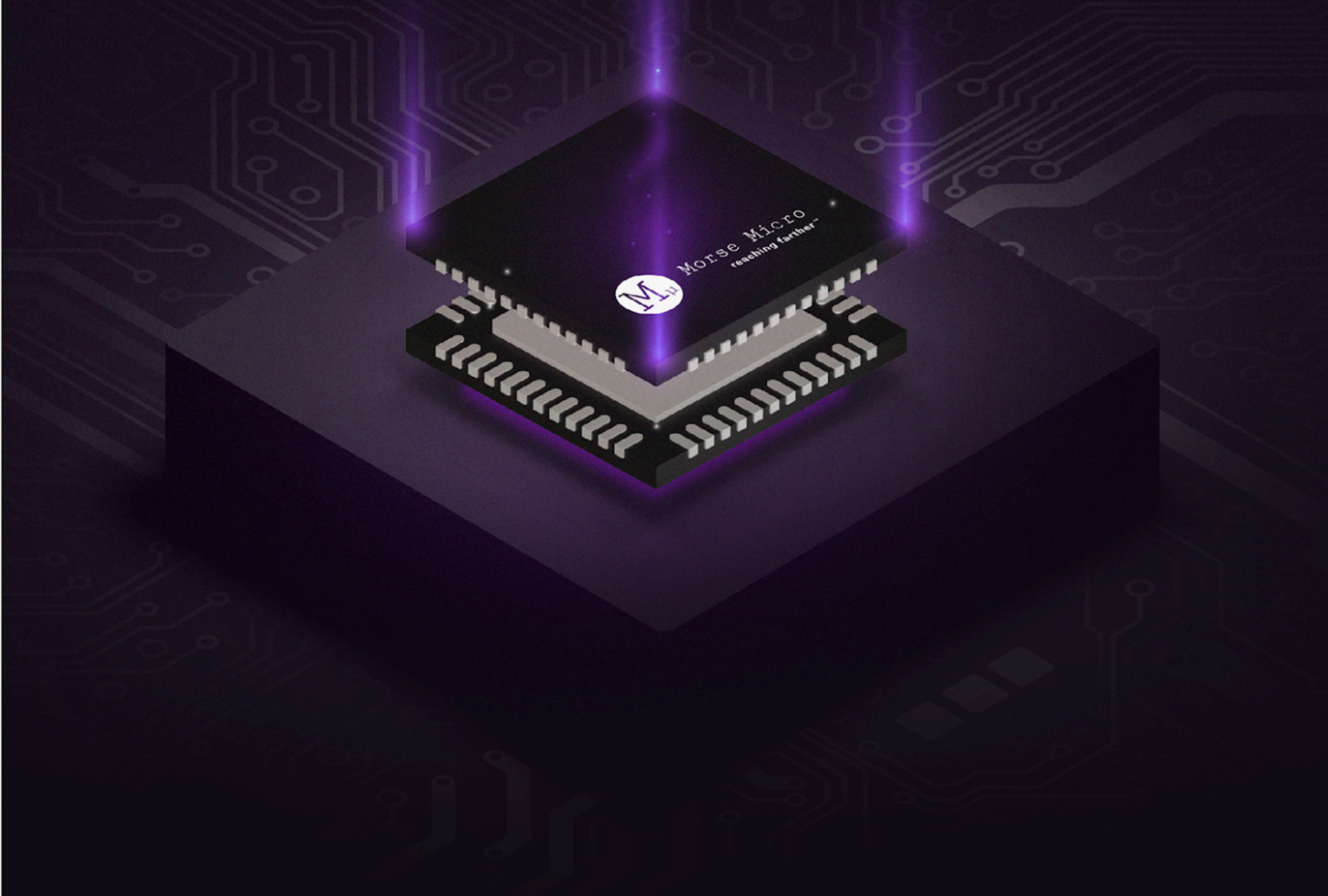
About Morse Micro
Morse Micro is a leading Wi-Fi HaLow fabless semiconductor company based in Sydney, with global offices. As the world's premier Wi-Fi HaLow company, we pioneer next-gen IoT wireless connectivity solutions. Morse Micro is now sampling its Wi-Fi CERTIFIED HaLow MM6108 production silicon: the fastest, smallest, lowest power and longest-range Wi-Fi HaLow chip available in the market.



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